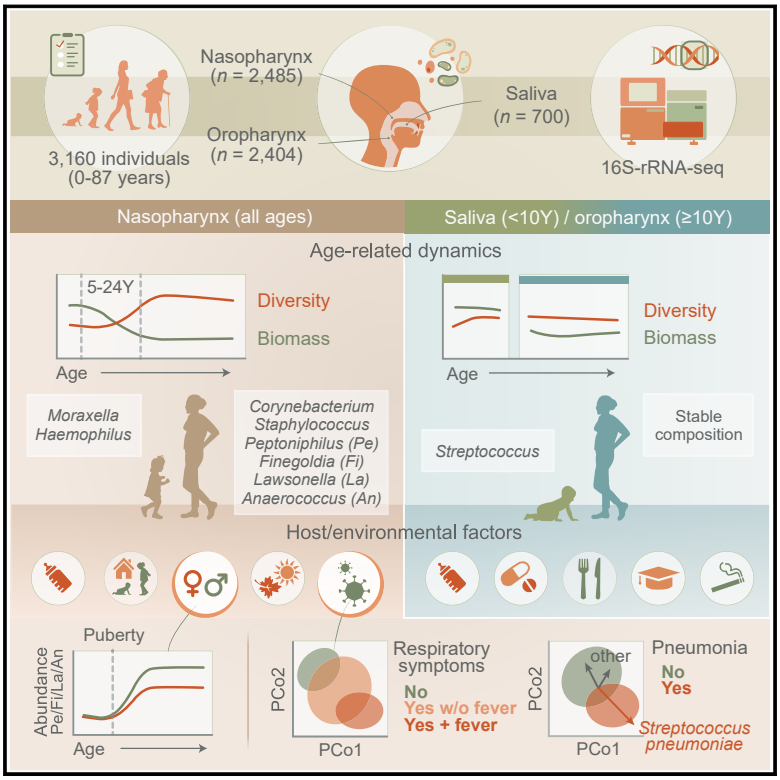
Cell

Host and environmental factors shape upper airway microbiota and respiratory health across the human lifespan

Graphical abstract



Highlights

- Atlas of nasopharyngeal (NP) and oral microbiota across the human lifespan
- Biomass, diversity, and composition of NP microbiota are strongly related to age
- Lifestyle factors are related to oral microbiota, and environment is related to NP microbiota
- NP microbiota associated with (recent) history of respiratory symptoms and pneumonia

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In brief

A large study of the nasopharyngeal and oral microbiota in the population of the Netherlands reveals associations between host and environmental factors, upper respiratory tract microbiota, and health outcomes that are strongly dependent on age and other factors.

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Host and environmental factors shape upper airway microbiota and respiratory health across the human lifespan

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SUMMARY

Our understanding of the normal variation in the upper respiratory tract (URT) microbiota across the human lifespan and how these relate to host, environment, and health is limited. We studied the microbiota of 3,104 saliva (<10 year-olds)/oropharynx (≥10 year-olds) and 2,485 nasopharynx samples of 3,160 Dutch individuals 0–87 years of age, participating in a cross-sectional population-wide study (PIENTER-3) using 16S-rRNA sequencing. The microbiota composition was strongly related to age, especially in the nasopharynx, with maturation occurring throughout childhood and adolescence. Clear niche- and age-specific associations were found between the microbiota composition and host/environmental factors and health outcomes. Among others, social interaction, sex, and season were associated with the nasopharyngeal microbial community. By contrast, the oral microbiota was more related to antibiotics, tobacco, and alcohol use. We present an atlas of the URT microbiota across the lifespan in association with environment and health, establishing a baseline for future research.

INTRODUCTION

Human health is strongly influenced by physiologically active microbial communities in and on the human body. Although less well studied compared with the gut microbiota, the upper respiratory tract (URT) microbiota have been associated with (respiratory) health outcomes, including susceptibility to and severity of respiratory infection,[1,2] chronic respiratory conditions such as asthma[3] and chronic obstructive pulmonary disease (COPD),[4] and cardiovascular diseases.[5] The respiratory tract is composed of multiple ecologically distinct niches, providing a plethora of habitats for diverse microbial communities. These microbial communities have a crucial protective role in the respiratory system by interacting with the mucosal immune system and providing a barrier against harmful microorganisms.[6–8] The URT microbiota is considered the gatekeeper to the lower airways and is thus important for preventing the entry of harmful pathogens and maintaining a healthy respiratory system.[6] The URT is directly exposed to the external environment and has to continuously accommodate changes in both host and environmental characteristics. Therefore, understanding the makeup of the URT microbiota in a general population and the impact of these factors on health is essential for improving our understanding of how the microbiota may affect or prevent acute and chronic respiratory diseases.

Thus far, research on the URT microbiota has been largely directed at infants and young children, which led to the identification of factors governing microbiota development, such as birth mode[9] and feeding type,[10,11] and has linked early life microbiome development to health outcomes.[12,13] Studies in young children at times of acute infections also found a clear link between URT microbiota and (severity of) lower respiratory tract infections.[2,14] Although we have some understanding of how the URT microbiota affects health in childhood, it is unclear how these relationships persist or change in adults. Limited research available has shown connections between the adult URT microbiota and factors such as social contact[15,16] and tobacco use.[17,18] Still, there is a lack of understanding of the natural

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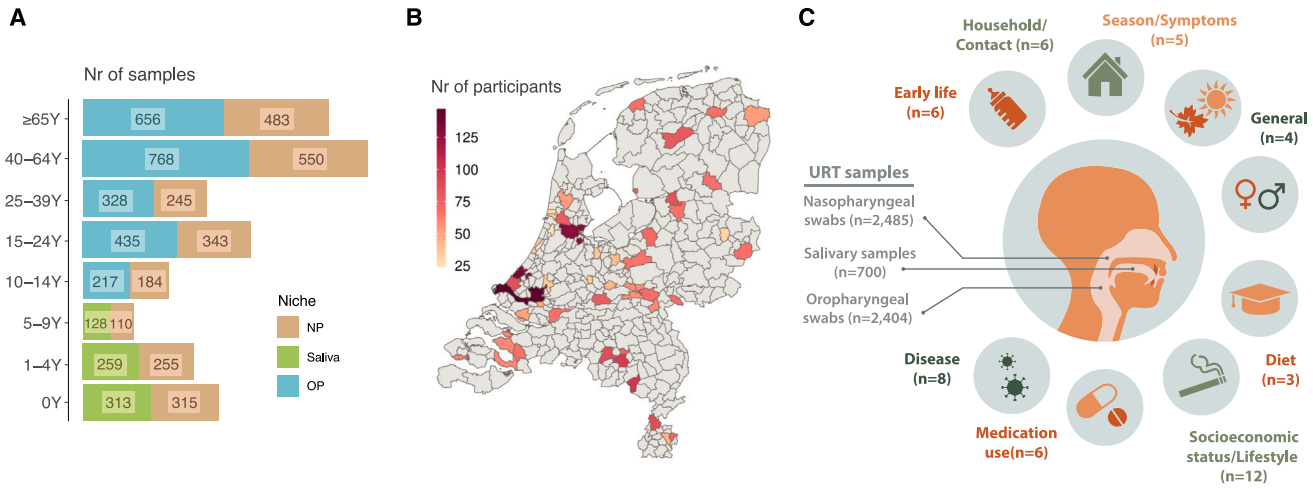


Figure 1. Overview of the sample population and data selection

(A) Number of eligible samples per age category.

(B) Geographical representation of the participants.

(C) The total number of analyzed samples per niche, and the number of selected variables per category from the questionnaires.

variation in the URT microbiota throughout the lifespan and the subsequent association with host, environmental, and health characteristics. Here, we studied the nasopharynx and oral samples of 3,160 individuals representing the Dutch population (0–87 year-olds), collected as part of a nationwide cross-sectional study from 2016/17 in the Netherlands.[19] We aimed to (1) characterize the URT microbiota of the general Dutch population across the lifespan and (2) its association with host, environmental, and health characteristics. Ultimately, our study can serve as a reference for future studies on the drivers of the natural variation in URT microbiota composition and how this is related to health outcomes.

RESULTS

Characteristics of the study population

We characterized the URT microbiota—nasopharynx and oral samples—from a Dutch population-based sample of 3,175 individuals in a cross-sectional fashion between January 2016 and December 2017. For practical reasons, we obtained saliva samples from individuals under 10 years of age and oropharynx samples from those aged 10 years and above, both representing the oral niche. Nasopharynx samples were obtained across all ages. After pre-processing and quality control, 5,589 URT samples from 3,160 individuals—3,104 oral and 2,485 nasopharynx—out of the 5,930 original samples (94%) were eligible for further analyses.

The study population included 319 infants (0 years: 10%), 611 children (1–14 years: 19%), and 2,230 adolescents/adults (Figure 1A; ≥15 years: 71%). We had matching samples from both the nasopharynx and oral cavity for 2,429 individuals (77%). Participants lived across the country (Figure 1B), were aged between 0 and 87 years (mean: 36 years), 51% of them were female, and 86% were born in the Netherlands (Table 1). In total, 82% reported their health to be good or very good (Table S1). Host, environmental, and health characteristics were categorized (Figure 1C; general, season/symptoms, household/contact, socioeconomic status/lifestyle, medication use, disease, diet, and early life).

The URT microbiota across the human lifespan

The dataset included 4,096 amplicon sequence variants (ASVs) with a relative abundance of ≥0.1% in at least 2 samples, covering 22 phyla, 40 classes, 96 orders, 176 families, 441 genera, and 660 species. To assess variation in overall community composition across the lifespan and differences between URT niches, we performed a principal coordinate analysis (PCoA). We observed a strong separation between nasopharynx and oral niches (saliva and oropharynx) with 30% of total variance (R^2) explained by niche (Figures 2A–2C; *PERMANOVA* p value < 0.001). Despite that saliva samples were only collected for individuals <10 years old, they roughly formed a continuum with oropharynx samples (Figure 2D). Age was the most important driver of microbial communities, explaining 10%, 11%, and 1% of variance in nasopharyngeal (all ages), salivary (<10 years), and oropharyngeal communities (≥10 years), respectively (Figures 2E and 2F; p values < 0.001). For the nasopharynx, we conducted two additional stratified PERMANOVA tests in ≤24 year-olds and >24 year-olds, showing that age explains substantially more variance in individuals aged 0–24 years compared with those older than 24 years (12% and 0.7%, respectively, p values < 0.001), underscoring age-related changes mostly occur up to adulthood. We identified 15 core genera (out of 392) (prevalence of >75%)[20] in the adult nasopharynx (40–64 years), constituting 94% of the total composition. These genera also accounted for 89%–95% of the overall microbial community composition in the other age categories, indicating early establishment of essential (core) microbes. However, several core genera including *Staphylococcus*, *Peptoniphilus*, *Finegoldia*, *Lawsonella*, and *Anaerococcus* only emerged from age 10

Table 1. Characteristics of the participants that were retained after pre-processing of the samples

	Age categories								
	0 years	1–4 years	5–9 years	10–14 years	15–24 years	25–39 years	40–64 years	≥65 years	All ages
	N = 319	*N* = 264	*N* = 128	*N* = 219	*N* = 444	*N* = 332	*N* = 784	*N* = 670	*N* = 3,160
Female	151 (47%)	133 (50%)	64 (50%)	111 (51%)	238 (54%)	173 (52%)	409 (52%)	319 (48%)	1,598 (51%)
Born in the Netherlands									
	315 (99%)	256 (97%)	110 (86%)	203 (93%)	422 (95%)	292 (88%)	608 (78%)	525 (78%)	2,731 (86%)
Level of education									
Low	43 (15%)	46 (18%)	15 (12%)	49 (24%)	169 (41%)	45 (15%)	271 (37%)	364 (56%)	1,002 (34%)
Middle	112 (38%)	105 (41%)	54 (45%)	73 (36%)	204 (49%)	125 (41%)	237 (33%)	130 (20%)	1,040 (35%)
High	138 (47%)	103 (41%)	51 (42%)	82 (40%)	44 (11%)	136 (44%)	220 (30%)	160 (24%)	934 (31%)
Smoking status									
Passive smoker (indoor)	N/A	N/A	N/A	N/A	36 (8.9%)	8 (2.7%)	26 (3.8%)	25 (3.9%)	126 (6.1%)
Past active smoker	N/A	N/A	N/A	N/A	22 (5.4%)	67 (22%)	270 (39%)	334 (52%)	693 (34%)
Active smoker	N/A	N/A	N/A	N/A	90 (22%)	58 (19%)	117 (17%)	65 (10%)	330 (16%)
Drinks alcohol									
	N/A	N/A	N/A	N/A	286 (69%)	241 (81%)	525 (75%)	476 (74%)	1,528 (74%)
Season of sampling									
Summer	58 (18%)	64 (24%)	38 (30%)	40 (18%)	89 (20%)	69 (21%)	195 (25%)	150 (22%)	703 (22%)
Fall	77 (24%)	70 (27%)	30 (23%)	77 (35%)	113 (25%)	90 (27%)	166 (21%)	158 (24%)	781 (25%)
Winter	101 (32%)	53 (20%)	18 (14%)	40 (18%)	119 (27%)	67 (20%)	165 (21%)	139 (21%)	702 (22%)
Spring	83 (26%)	77 (29%)	42 (33%)	62 (28%)	123 (28%)	106 (32%)	258 (33%)	221 (33%)	972 (31%)
Respiratory symptoms in the last month									
Runny nose	206 (66%)	181 (69%)	53 (43%)	105 (49%)	276 (64%)	174 (53%)	276 (36%)	228 (34%)	1,499 (48%)
Cough	140 (45%)	128 (49%)	33 (26%)	55 (25%)	161 (38%)	84 (26%)	188 (24%)	164 (25%)	953 (31%)
Fever	114 (36%)	93 (36%)	17 (14%)	12 (5.6%)	42 (9.9%)	29 (9.0%)	48 (6.3%)	34 (5.2%)	389 (13%)
Suffered from pneumonia in last 3 years									
One pneumonia	N/A	N/A	N/A	N/A	11 (2.5%)	17 (5.3%)	45 (5.9%)	49 (7.4%)	122 (5.6%)
Multiple pneumonia	N/A	N/A	N/A	N/A	8 (1.8%)	8 (2.5%)	22 (2.9%)	21 (3.2%)	59 (2.7%)
Used antibiotics in last 3 months									
	22 (9.4%)	21 (11%)	5 (5.3%)	7 (4.3%)	26 (9.1%)	18 (8.3%)	68 (12%)	93 (17%)	260 (11%)
Contact during work									
Animals	N/A	N/A	N/A	N/A	30 (11%)	21 (9.7%)	62 (11%)	42 (7.7%)	155 (9.6%)
Children	N/A	N/A	N/A	N/A	59 (21%)	46 (21%)	105 (18%)	81 (15%)	291 (18%)
Nursing home residents	N/A	N/A	N/A	N/A	35 (12%)	24 (11%)	91 (16%)	74 (14%)	224 (14%)
Patients	N/A	N/A	N/A	N/A	40 (14%)	35 (16%)	72 (13%)	14 (2.6%)	161 (10.0%)
Lives with children younger than 5 years									
	108 (38%)	66 (28%)	46 (42%)	17 (9.0%)	17 (4.3%)	100 (39%)	32 (5.3%)	0 (0%)	386 (16%)

More extensive information depicted in Table S1.

onward (Figure S1A). For the adult oropharynx, we identified 22 core genera (out of 115), representing 96%–97% of the total oropharyngeal community composition and also representing 92%–97% of the communities in saliva samples (Figure S1B). Nasopharynx samples showed higher heterogeneity (mean Bray-Curtis dissimilarity $\bar{x}$ = 0.76) than saliva ($\bar{x}$ = 0.53) and oropharynx ($\bar{x}$ = 0.63) samples (Figure 2G), especially in older individuals (Figure S1C). In most age categories, the concordance between matching samples within individuals was higher compared with the concordance between non-matched nasopharynx and oral samples, underscoring ecological continuity between niches within individuals. Infants exhibited the greatest concordance among matched samples, indicating a lower degree of niche differentiation (Figure S1D; $\bar{x}$ = 0.91; *q* value = 0.01).

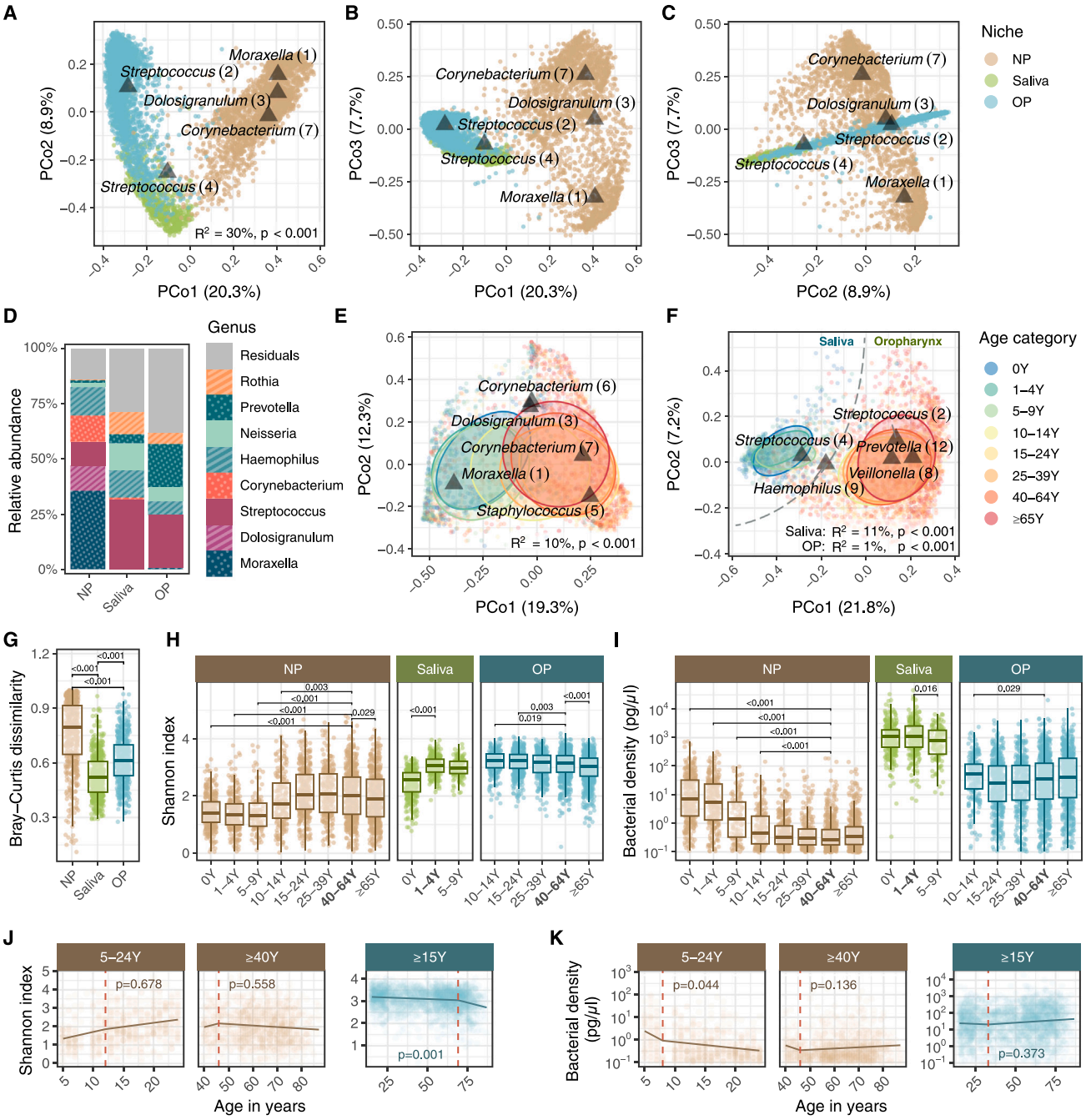


Figure 2. The upper respiratory tract microbiota across ages

(A–C) Principal coordinate analysis (PCoA) based on Bray-Curtis dissimilarities of all the samples, colored by niche. The variance explained by the first three principal coordinates is depicted as percentages in brackets. The five highest-ranked ASVs were simultaneously visualized as triangles. Explained variance (R^2) and p values were calculated using PERMANOVA tests.

(D) Average relative abundance of the most abundant genera per niche (average relative abundance >10%) in individuals aged 7–10 years.

(E and F) PCoA based on Bray-Curtis dissimilarities stratified by niche colored by age category (E, nasopharynx [NP]; F, saliva/oropharynx [OP]). Explained variance (R^2) and p values were calculated using PERMANOVA tests.

(G) Bray-Curtis dissimilarities between samples per niche in individuals aged 7–10 years (sub-sampling n = 500 per niche). Boxplots represent the 25th and 75th percentiles (lower and upper boundaries of boxes, respectively), the median (middle horizontal line), and measurements that fall within 1.5 times the interquartile range (IQR; distance between the 25th and 75th percentiles; whiskers). p values were calculated using linear models and corrected for multiple testing with the Benjamini-Hochberg (BH) method (q values).

(legend continued on next page)

Next, we explored the within-sample (alpha-)diversity per niche, showing a strong association between age and diversity for both the nasopharynx and oral niches (Figure 2H). In the nasopharynx, we found stable Shannon indices across the age categories 0, 1–4, and 5–9 years (all comparisons; q value > 0.05), after which a stepwise increase in diversity was seen in 10–14 year-olds (β = 0.4; q value < 0.001; reference 5–9 years) and 15–24 year-old (Table S3; β = 0.3; q value < 0.001; reference 10–14 years), indicating that nasopharyngeal microbiota continues to develop until young adulthood (15–24 years). By contrast, we found that the overall bacterial density strongly decreased with age, especially in 5–14 year-olds (Figure 2I; Table S3; q value < 0.05). Using piecewise regression, we examined age as a continuous variable within individuals aged 5–24 years, and 40 years and older to identify potential breakpoints within these age ranges. In individuals aged 5–24 years, we observed a more pronounced decrease in bacterial density among 5–8 year-olds, followed by a slightly attenuated decline (p value = 0.044). Other comparisons showed no significant slope differences before and after breakpoints, suggesting a gradual change in biomass and diversity across the age ranges (Figures 2J and 2K).

In contrast to the nasopharynx, we found a strong increase in diversity in saliva between 0 and 1–4 year-olds (Figure 2H; β = 0.6; q value < 0.001). Upon closer examination of infants (0 years), increase in diversity was observed up until the fifth month of life (Figure S1E; q value < 0.001). The oropharyngeal microbiota showed a gradual decline in microbial diversity across adulthood, with a significantly lower diversity in elderly participants (≥65 years) compared with adults (40–64 years) (Figure 2H; β = −0.132; q value < 0.001). Piecewise regression analyses revealed a steady decline in microbial diversity in the oropharynx until the age of 69, after which the decrease became more pronounced (Figures 2J and 2K; p value = 0.001). Bacterial density in the oral niches remained relatively constant across ages (Figures 2I, 2K, and S1F).

Microbial cluster and ASV dynamics across ages

To gain insight into the microbial community structure in both nasopharynx and oral niches across the age categories, we performed complete linkage hierarchical clustering on Bray-Curtis dissimilarity matrices. We identified ten microbial clusters within the nasopharynx, generally typified by the abundance of a single or few ASVs (Figure S2A; Table S4). Within the oral niches, we identified eight clusters, with two microbial clusters dominated by *Streptococcus* (2 and 4), and six microbial clusters represented by mixtures of ASVs (Figure S2B; Table S4).

Next, we used logistic regression to examine age-specific patterns, using age in years as the predictor and cluster membership as the outcome, adjusting for sex and season of sampling. We observed a negative association between age and four nasopharyngeal microbial clusters characterized by high relative abundance of *Moraxella* (1 and 17) and *Haemophilus* (10 and 19) (Mor1, Mor17, Hae10, Hae19), indicating that these clusters are more frequent in younger individuals (OR < 1; q values < 0.001). By contrast, the four clusters dominated by *Corynebacterium* (6 and 7), *Dolosigranulum pigrum* (3) and *Staphylococcus* (5) (Cor6, Cor7, Dol/Cor, Sta5) were positively associated with older ages (Figures 3A and 3B; OR > 1; q values < 0.001). When comparing the age categories (i.e. 0, 1–9, 10–24, 25–64 [reference], and ≥65 year-olds), we observed the most pronounced differences in 0 and 1–9 year-olds compared with 25–64 year-olds, with notably higher membership in nasopharyngeal clusters Mor1, Mor17, and Hae19 and lower membership in clusters Sta5 and Dol/Cor among younger participants (Figure 3C). Additionally, we studied the dynamics of the absolute abundance of the most abundant ASVs by multiplying the total bacterial density (measured by quantitative PCR) with the relative abundance for each ASV per sample,[21] and performed linear models. These analyses showed a higher absolute abundance of *D. pigrum* (3) and *Corynebacterium* (6) in the nasopharynx of <10 year-olds compared with those 40–64 years of age, despite the fact that these clusters were more prevalent in adults (Figure S3A; Table S5; q value < 0.05). In addition, we identified strong associations between the nasopharyngeal ASVs *Anaerococcus octavius* (66), *Anaerococcus* (97), *Corynebacterium* (46), *Finegoldia magna* (39), *Lawsonella clevelandensis* (45), and *Peptoniphilus* (29) and older age. Similar to the microbial diversity and bacterial density results, the absolute abundance of the aforementioned ASVs increased until the age of 15–24 years, after which they leveled off (Figure S3A).

Regarding oral samples, we observed four microbial clusters in saliva (<10 year-olds). We found that *Streptococcus*-dominated clusters were predominant in infants (0 years) (Str4: 81.8%; Str2: 3.5%), though they were significantly lower in 1–4 followed by 5–9 year-olds (Figure 3D). The third largest cluster, the *Neisseria* (13) and *Haemophilus* (9)-dominated cluster (Nei13/Hae9), showed a relative increase from infants to older children (OR > 1; q value < 0.001). On ASV level, we confirmed an increase in *Haemophilus* (9 and 10) and *Neisseria* (13) with age in infants (Figure S3C). In oropharynx samples (≥10 years), we observed a relatively stable distribution of oral clusters across age. The *Streptococcus* (2) dominated cluster was, after the initial drop in early life, positively associated with older ages (OR > 1; q value < 0.001; 19.8% in 10–14 year-olds and 31.6% in ≥65 year-olds). Stratified analyses confirmed that the cluster membership of the Str2 cluster was significantly higher in ≥65 year-olds compared with 25–64 year-olds (Figure 3F; q value = 0.002). Additionally, we found that three clusters were negatively

(H and I) ASV-level Shannon diversity (rarefied) (H) and bacterial density (I) across age, stratified by niche. Boxplots represent the 25th and 75th percentiles, the median, and measurements that fall within 1.5 times the interquartile range. p values were calculated using linear models and corrected for multiple testing with the BH method (q values).

(J and K) ASV-level Shannon diversity (J) and bacterial density (K) across age in years in three age categories for the nasopharyngeal (5–24 years, ≥40 years) and oropharyngeal (≥15 years) microbiota. Through piecewise regression analysis and the Davies test, significant breakpoints (denoted by red dashed lines) were identified, indicating shifts in Shannon diversity or bacterial density at specific age points.

All models were adjusted for sex and season of sampling. The reference groups are indicated in bold.

See also Figure S1.

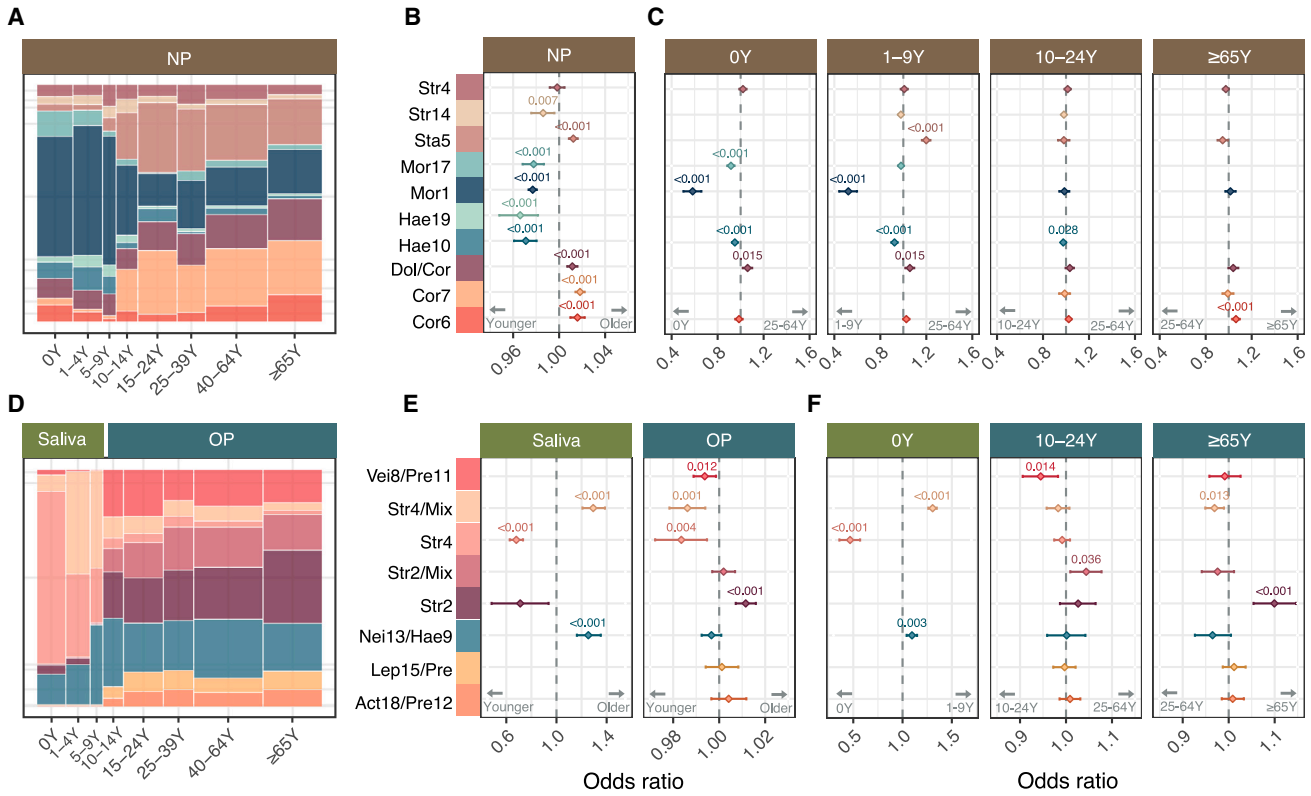


Figure 3. The upper respiratory tract microbial clusters

(A and D) Mosaic plot of the proportion of participants in each nasopharyngeal (A) and oral (D) microbial cluster per age category. Cluster names are based on the most dominant ASVs in each cluster. Nasopharynx clusters were driven by *Moraxella* (1) (Mor1; 28% of samples), *Staphylococcus* (5) (Sta5; 18%), *Corynebacterium* (7) (Cor7; 18%), *Corynebacterium* (6 and 7)/*Dolosigranulum pigrum* (3) (Cor/Dol; 13%), *Corynebacterium* (6) (Cor6; 6%), *Streptococcus* (4) (Str4; 5%), *Haemophilus* (10) (Hae10; 3.8%), *Moraxella lincolnii* (17) (Mor17; 3.5%), *Streptococcus* (14) (Str14; 2.5%), and *Haemophilus* (19) (Hae19; 1.5%). Oral clusters were dominated by *Streptococcus* (2) (Str2; 19.1%), *Streptococcus* (4) (Str4; 13.6%), mixtures of microbes with *Streptococcus* (2) (Str2/Mix; 12%), *Streptococcus* (4) (Str4/Mix; 11%), or combinations of *Veillonella* (8)/*Prevotella* (11) (Vei8/Pre11; 13%), *Neisseria* (13)/*Haemophilus* (9) (Nei13/Hae9; 22%), *Leptotrichia* (15)/*Prevotella*, (Lep15/Pre; 5%), and *Actinomyces* (18)/*Prevotella* (12) (Act18/Pre12; 5%).

(B and E) Odds ratio determined by logistic regression for the correlation between age in years and cluster membership of nasopharyngeal (B) and oral (E) clusters across all ages. The horizontal lines connecting the circles illustrates the lower and upper bounds of the 95% confidence interval. *p* values were corrected for multiple testing using the Benjamini-Hochberg (BH) method (*q* values).

(C and F) Logistic regression-derived odds ratios for age categories and cluster membership in nasopharyngeal (C) and oral microbiota (F). The horizontal lines connecting the circles illustrates the lower and upper bounds of the 95% confidence interval. *p* values were corrected for multiple testing using the BH method (*q* values). Reference groups: nasopharynx and oropharynx, 25–64 years; saliva, 1–9 years.

All models were adjusted for sex and season of sampling.

See also Figures S2 and S3.

associated with age (Figures 3D–3E; Vei8/Pre11, Str4, and Str4/Mix, OR < 1; *q* value < 0.05), with Str4 cluster membership higher in the age category 0 years compared with 1–9 years (Figure 3F). Using differential abundance analysis, we confirmed a gradual decrease in absolute abundance from 10 years onward for numerous ASVs including *Haemophilus* (10), *Alloprevotella* (49 and 65), *Actinobacillus* (52), *Pasteurellaceae* (77), and *Veillonella massiliensis* (78) (Figure S3B), which was in line with the Shannon diversity results.

Association between URT microbiota with host, environmental, and health characteristics

To determine how much host/environmental factors and health outcomes contribute to the variation in the URT microbial community composition, we performed univariable PERMANOVA tests on the separate age strata (Figure 4). The strongest association was observed for season of sampling and the nasopharyngeal microbiota composition in 6/8 age categories (R^2 = 1.2%–5.6%). The association between season and the nasopharyngeal microbiota was in line with recent symptoms (runny nose, cough, or fever at least one month prior to sampling), which occurred more frequently in infants (0 year-olds; R^2 = 1.3%–1.6%). Similarly, eating solid food (R^2 = 2.0%) and day-care attendance (R^2 = 1.4%) were strongly associated with microbial variation in infants. Additionally, our findings showed an association between household composition, specifically living with children under the age of 5 years, and the nasopharyngeal microbiota composition in both young children and participants aged 25–39 years (R^2 = 1.7%), with the latter most likely to be parents. For the oropharynx, we found that socioeconomic status/lifestyle, recent antibiotic use,

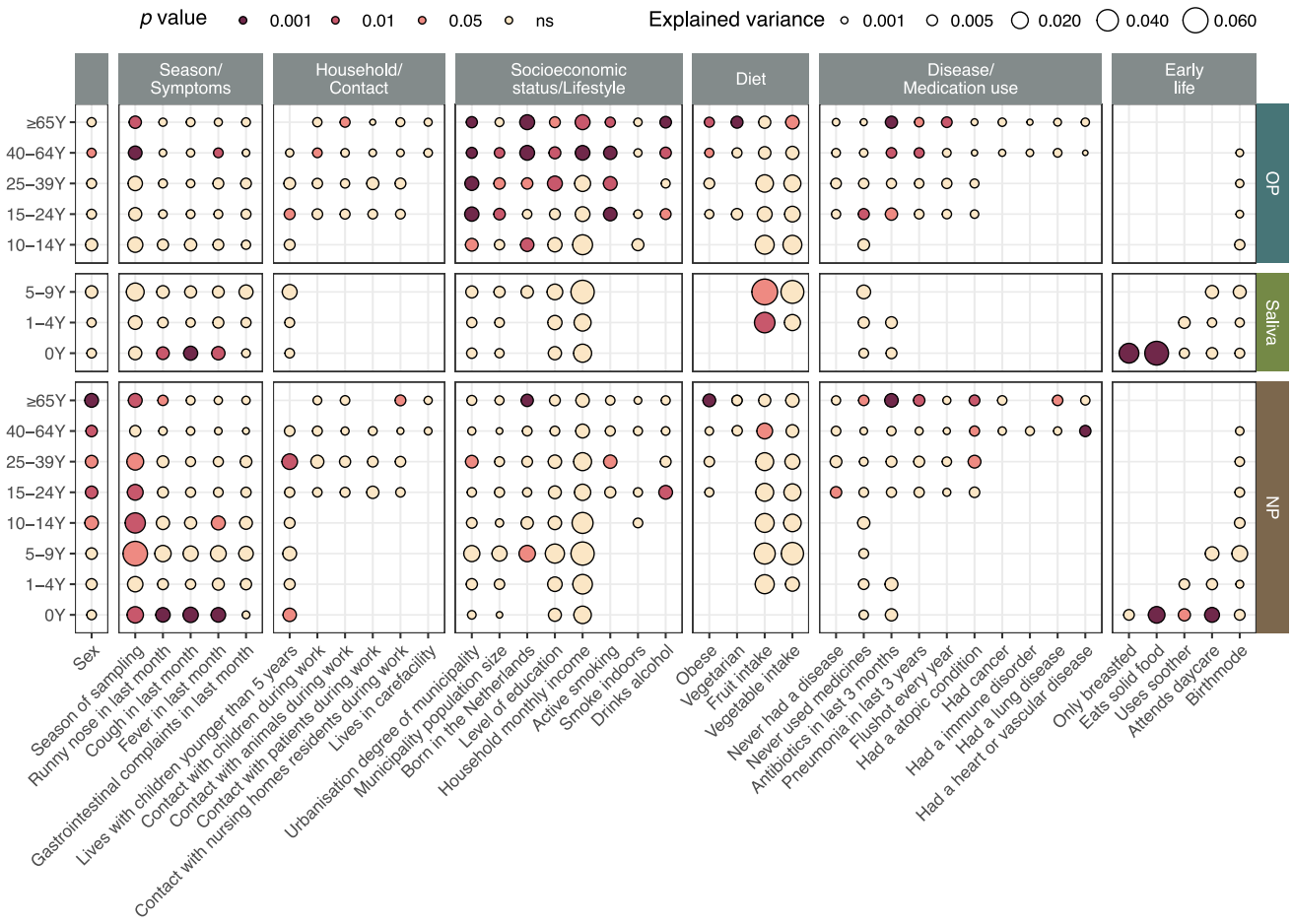


Figure 4. Age-stratified associations between host/environmental/health characteristics and the upper respiratory tract microbial community composition

Microbiota variance (R^2) explained by host, environmental, and health characteristics, and *p* values as determined by univariable PERMANOVA tests across age and per niche. Associations were only tested when there were at least 10 samples per group. The color indicates the level of significance. The circle size indicates the explained variance. ns, not significant.

See also Figure S4.

and reports on pneumonia within the last 3 years were strongly associated with microbiota composition, especially in older individuals (≥40 years) (Figure 4). Among others, a substantial influence of urbanization degree (R^2 = 0.5%–4.0%) and municipality size (R^2 = 0.2%–0.6%) on the URT microbiota composition was observed across most age categories, particularly in participants aged 15–64 years. Diet, especially fruit intake (R^2 = 4.0%–6.0%), had a significant influence on the salivary microbiota composition. We also found that feeding type in infants (exclusively breastfed R^2 = 3.2%; eats solid food R^2 = 5.3%; p value < 0.001) had a more pronounced influence on the salivary than on the nasopharynx microbiota (Figure 4). Again, recent symptoms of respiratory infections were most strongly associated with the oral microbiota of infants compared with the other age categories.

To gain a deeper understanding of these patterns, we examined the relationship of host, environmental, and health characteristics with the URT microbiota in terms of (1) diversity, (2) density, and (3) differential abundance on ASV/genus level. We used linear models (MaAsLin2) to analyze the nasopharynx at the ASV level and the oral niches at the genus level. Our goal was to identify the specific taxa responsible for the observed variations. For the oral niches, we chose to zoom in on genus rather than ASV level, as many ASVs from the same genera had similar responses in previous analyses, and this approach allows for a broader representation of the oral microbiota. We focused on the top 31 ASVs and 27 genera for the nasopharynx and oral niches, respectively (all representing >1% relative abundance in at least 100 samples), together covering 83% of the relative abundance in nasopharynx samples and 98% in oral niches. For each model, we corrected for age, sex, and season of sampling. Additional corrections were applied based on the PERMANOVA results and degree of correlation (Table S2).

Associations between URT microbiota and sex

First, we investigated sex-related trends in the URT microbiota, prompted by the links discovered between the gut microbiota

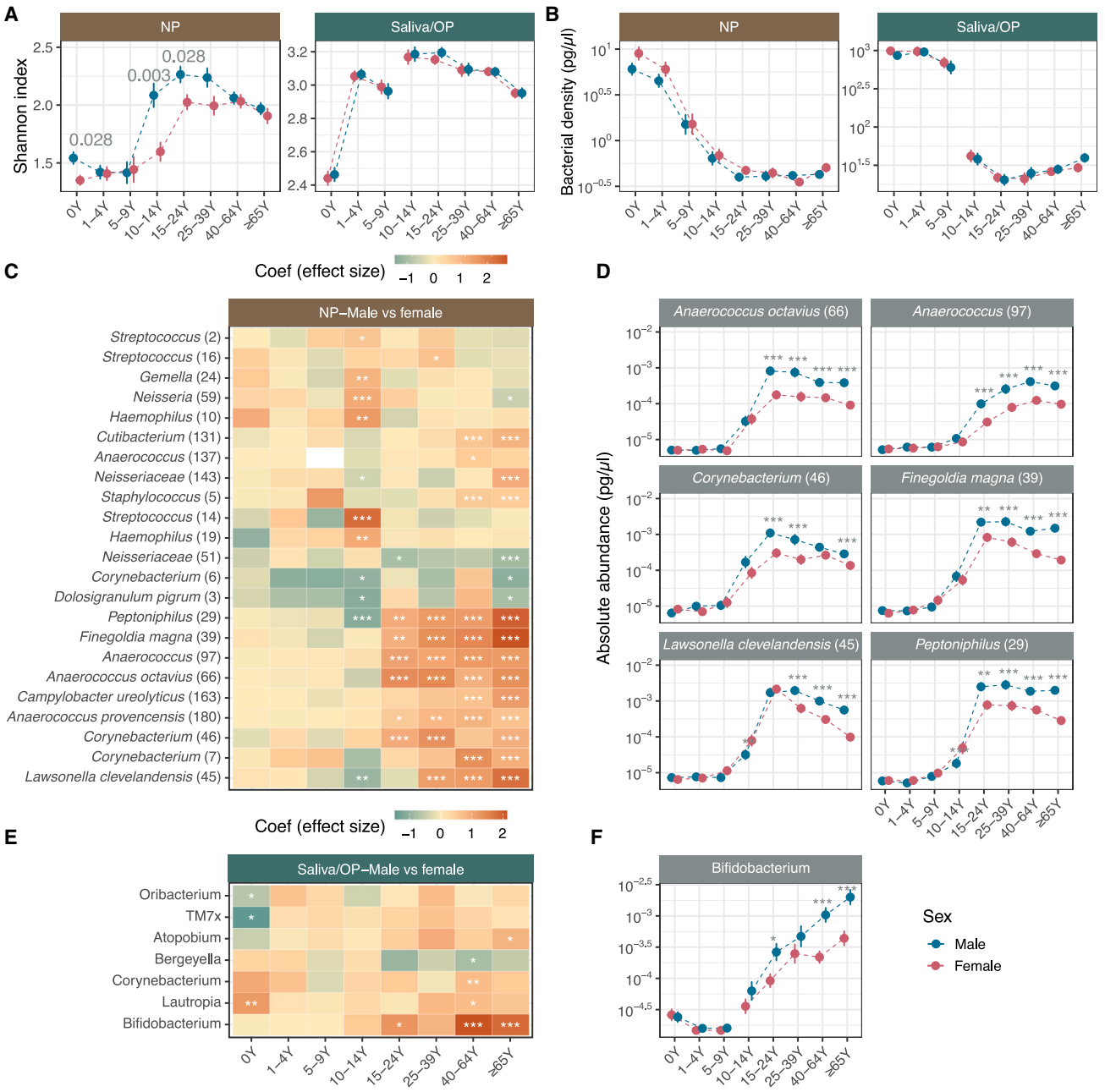


Figure 5. Sex-related patterns in the upper respiratory tract microbiota

(A and B) The relationship of Shannon diversity (A) and bacterial density (B) (outcome) with sex per age category. *p* values were calculated using linear models and corrected for multiple testing with the Benjamin-Hochberg (BH) method (*q* values). Plots visualize the mean and standard error.

(C) Associations between the absolute abundance of ASVs and sex in the nasopharynx (NP) across the lifespan. Estimates and *q* values were calculated using MaAsLin2. The plot shows all the ASVs with at least one significant comparison. The color corresponds to the direction of the association (red for positive, green for negative).

(D) Absolute abundance of ASVs in the nasopharynx with a significant response to sex (*q* value < 0.1) in at least three age categories. Plots visualize the mean and standard error.

(E) Associations between the absolute abundance of the genera and sex in the oral niches. Estimates and *q* values were calculated using MaAsLin2. The plot shows all the ASVs with at least one significant comparison. The color corresponds to the direction of the association (red for positive, green for negative).

(F) Genera with at least one significant (*q* value < 0.1) response to sex in the oral niches across age. Plots visualize the mean and standard error.

q value ≤ 0.25, **q* value ≤ 0.1, ***q* value ≤ 0.05. Models were corrected for age in years/months and season of sampling.

and hormonal disparities between males and females.[22] On a community level, we showed that sex explained a modest 0.3% of the total variance in the nasopharynx (Table S6; multivariable PERMANOVA p value < 0.001). We performed linear models per age category and detected a higher microbial diversity in the nasopharyngeal microbiota of males between the ages of 10–24 years and in infants 0 years compared with females of the same age (Figures 5A and 5B; q values < 0.05). In males 15 years and older, we observed higher levels of several low-abundant ASVs, including *A. octavius* (66), *Anaerococcus* (97), *Corynebacterium* (46), *F. magna* (39), *L. clevelandensis* (45), and *Peptoniphilus* (29). (Figures 5C and 5D; q values < 0.1 in at least 3 age categories). Interestingly, these ASVs were also independently associated with the older ages mentioned previously (Figure S3A). We performed sensitivity analysis using multivariable linear regression models and eliminated the possibility that sex-related effects were solely influenced by alterations in lifestyle factors, including smoking and drinking (data not shown). For the microbiota of both oral niches, we observed a similar modest effect of 0.1% variance explained by sex, although this was only significant for oropharynx samples (Figure 4; Table S6; p value = 0.002). Our results showed that *Bifidobacterium* was particularly abundant in the oropharynx of older males (≥40 years) (Figures 5E and 5F; q value < 0.1).

Associations between URT microbiota and social contacts

Next, because understanding the potential transmission of microbes through personal contact is crucial for effective disease prevention and management,[15] we examined the impact of household composition and work-related contacts (work with children, animals, patients, or elderly) on the URT microbiota of adults aged 25–64 years. To extract independent signals for each contact type, each model was adjusted for all other contact variables. We observed particularly strong signals associated with the nasopharyngeal microbiota that could be attributed to personal contact with children in the household (Table S6; R^2 = 1.2%; p value < 0.001) and work environment (R^2 = 0.6%; p value = 0.005). In adults that live or work with children, we found a lower microbial diversity, a higher bacterial density, and a higher absolute abundance of *Moraxella* (1 and 17) and *D. pigrum* (3) (Figures S4A–S4C), which is typical for younger children (Figures 2H, 2I, and S3A). By comparing the Bray-Curtis dissimilarities, we confirmed that adults who had contact with children had a nasopharyngeal microbiota that closely resembled that of a child (Figure S4D). By contrast, the oropharyngeal microbiota was less affected by living (R^2 = 0.3%; p value = 0.029) or working (R^2 = 0.2%; p value = 0.019) with children. Therefore, our results suggest that contact with children plays a larger role in shaping the adult nasopharyngeal microbiota compared with the oropharyngeal microbiota.

Associations between URT microbiota with tobacco and alcohol use

Because tobacco use is associated with respiratory tract infections and chronic respiratory disease, we studied the association between tobacco use and the URT microbiota in individuals aged 15 years or older, including active smokers (16%), former smokers (33.6%), and passive smokers (6.1%). We corrected for education level (*Pearson r* = −0.09; q value < 0.001) and alcohol consumption (Table S2; *Pearson r* = 0.09; q value < 0.001). In the nasopharyngeal microbiota, 0.3% of the variation was explained by smoking status (Table S6; p value < 0.001). Among others, we observed a stepwise reduction in the absolute abundance of *D. pigrum* (3) from absence of exposure to individuals with a history of active smoking, to passive smokers, to current smokers (Figure S4G; q value < 0.25). Conversely, rare taxa, including *Peptoniphilus* (29), *F. magna* (39), *Corynebacterium* (46), and *Campylobacter ureolyticus* (163), were more abundant in active smokers compared with never smokers (q value < 0.1), with *Peptoniphilus* (29) and *F. magna* (39) positively associated with the number of cigarettes smoked per week (Figure S4H). A higher degree of microbiota variation could be attributed to active smoking in the oropharynx compared with the nasopharynx (Table S6; R^2 = 0.6%; p value < 0.001). Here, the microbial diversity in past and current smokers was lower compared with never smokers (Figures S4E and S4F; q value < 0.05). The oropharyngeal microbiota of active smokers had a higher absolute abundance of *Bifidobacterium* (coef = 3.0; q value < 0.05)[18] and a lower absolute abundance of 17 other genera, particularly *Neisseria* (Figure S4G; coef = −4.0; q value < 0.05). A sensitivity analysis confirmed that effects of smoking were independent of sex.

We additionally examined the association between alcohol consumption (Table S1; 25.7% abstained from alcohol consumption) and the URT microbiota, controlling for education (Table S2; Pearson r = 0.2; q value < 0.001) and active smoking. The effect of alcohol consumption was more apparent in the oropharynx (R^2 = 0.3%; p value < 0.001) compared with the nasopharynx (Table S6; R^2 = 0.1%; p value = 0.021). Among others, the absolute abundance of *Haemophilus* (9) (coef = 0.8; q value < 0.05) and *Streptococcus* (4) (coef = 0.7; q value < 0.05) was higher in the nasopharynx of individuals that consume alcohol compared with non-drinkers (Figure S4G). In the oropharynx, we observed a higher microbial diversity (q value = 0.005) and a higher absolute abundance of 14 genera, including TM7x, in individuals that consume alcohol compared with non-drinkers (Figures S4E and S4G; q values < 0.25). These results demonstrate a (different) impact of both tobacco use and alcohol consumption on the URT microbiota structure.

Associations between URT microbiota and antibiotic use

Antibiotics are generally used to eliminate bacterial pathogens from any organ system; however, antibiotic use likely also results in the disruption of the microbiome structure throughout the body.[23] We investigated the potential impact of antibiotic use on URT microbiota in individuals aged 15 years and older, as antibiotic usage was more prevalent in this age group (12.7% used antibiotics in the last three months). Antibiotic use within the last three months explained 0.2% and 0.4% of the variance in the nasopharyngeal (p value = 0.011) and oropharyngeal (p value < 0.001) microbiota, respectively (Table S6). In the nasopharynx, we found lower microbial diversity with antibiotic use (Figure S4E; q value = 0.045), accompanied by a lower absolute abundance of *D. pigrum* (3), *Peptoniphilus* (29), and *L. clevelandensis* (45) (Figure S4G; q values < 0.05). Similarly,

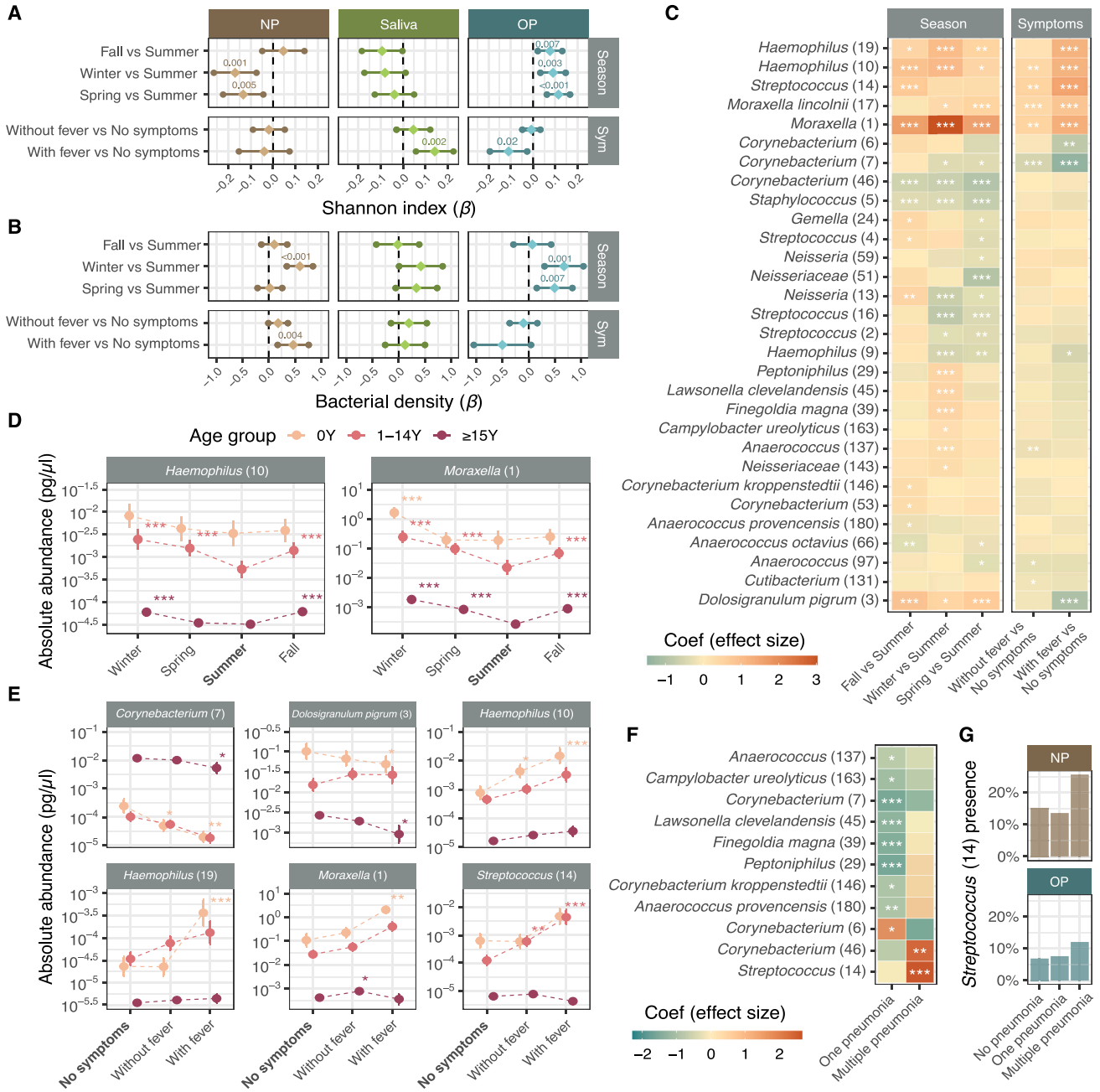


Figure 6. Season, respiratory symptoms, and pneumonia-related patterns in the upper respiratory tract microbiota

(A and B) Relationship of Shannon diversity (A) and bacterial density (B) (outcome) with season of sampling (adjusted for age category and sex) or respiratory symptoms (adjusted for age category, sex, and season of sampling), stratified by niche. The horizontal lines connecting the circles illustrates the lower and upper bounds of the 95% confidence interval. *p* values were calculated using linear models and corrected for multiple testing with the Benjamini-Hochberg (BH) method (*q* values).

(C) Associations between the absolute abundance of the ASVs with season of sampling and respiratory symptoms in the nasopharynx (NP). Estimates and *q* values were calculated using MaAsLin2. The plot shows all the ASVs with at least one significant comparison. The color corresponds to the direction of the association (red for positive, green for negative).

(D and E) ASVs with the greatest association (coef < −0.9 or coef > 0.9; *q* value < 0.05) with season of sampling (D) and respiratory symptoms (E) in NP stratified by the three age groups 0, 1–14, and ≥15 years. Estimates and *q* values were calculated using MaAsLin2. Plots visualize the mean and standard error. Asterisks indicate the significance of the concerning group versus the reference group depicted in bold.

(F) Associations between absolute abundance of the ASVs and history of pneumonia in NP (≥15 years) (adjusted for age category, sex, season of sampling, and antibiotic use). Estimates and *q* values were calculated using MaAsLin2. The plot shows all the ASVs with at least one significant comparison. The color corresponds to the direction of the association (red for positive, green for negative).

(legend continued on next page)

in the oropharynx, we observed significant lower microbial diversity (q value < 0.001), bacterial density (q value = 0.034), and absolute abundance of 19 genera (q values < 0.25) with antibiotic use (Figures S4E–S4G). Findings were similar in young children (<5 years), with lower absolute abundances of *D. pigrum* (3) (coef = −3.0; q value = 0.007) and *Corynebacterium* (7) (coef = −1.8; q value = 0.13) in the nasopharynx. Our findings confirmed that antibiotic use leads to changes in the microbiota of both URT niches, though more substantial in the oropharynx.

Associations between URT microbiota with season, respiratory symptoms, and pneumonia

To explore the relationship between season of sampling, the presence of respiratory symptoms, and the URT microbiota, we assessed microbiota differences between participants with or without respiratory symptoms (i.e., a runny nose and/or a cough at least one month prior to sampling) or with respiratory symptoms combined with fever. In our dataset, the four seasons were represented as follows: winter (22%), spring (31%), summer (22%), and fall (25%), whereas respiratory symptoms were more common in individuals younger than 5 years (respiratory symptoms without a fever: 39%; respiratory symptoms with a fever: 32%) and in the winter season (respiratory symptoms without a fever: 48%; respiratory symptoms with a fever: 17%). Season was found to account for 0.4% of the variance in the nasopharyngeal microbiota (Table S6; p value < 0.001). Deeper analysis of the nasopharynx revealed a reduced Shannon diversity (Figure 6A; q value = 0.005) and a higher bacterial density (Figure 6B; q value < 0.001) during the winter season compared with the summer season. Our findings showed that *Moraxella* (1) and *Haemophilus* (10 and 19) had a higher absolute abundance in the winter compared with the summer (Figure 6C; q values < 0.05). We analyzed seasonal patterns in different age groups (0, 1–14, and ≥15 year-olds) and demonstrated that seasonal fluctuation in *Haemophilus* (10) abundance is particularly visible in children (1–14 years) and adolescents/adults (≥15 years) (Figure 6D). In the salivary (<10 years) and oropharyngeal (≥10 years) microbiota, season of sampling explained 0.9% and 0.5% of the variance, respectively (p value < 0.001). In the oropharyngeal microbiota, the microbial diversity (Figure 6A; q value = 0.003), bacterial density (Figure 6B; q value < 0.001), and absolute abundance of 15 genera (Figure S5A; q values < 0.25) were higher in winter and spring compared with summer.

Although individuals with a recent respiratory infection generally reported mild symptoms, likely reflecting common intercurrent viral infections, these were still strongly associated with the nasopharyngeal microbiota composition (Table S6; R^2 = 1.7%; p value < 0.001). For the nasopharynx, we detected an overall higher bacterial density (Figure 6B; q value = 0.004), as well as a higher absolute abundance of known potential pathogens in individuals with recent respiratory symptoms, especially in those who had suffered from infections accompanied by fever (Figure 6C; q value < 0.05). These potential pathogens included *Haemophilus* (10 and 19), *Moraxella* (1 and 17), and *Streptococcus* (14). Based on BLASTn-alignment, we suspected *Streptococcus* (14) to reflect *Streptococcus pneumoniae* (Table S7; identity = 100%; E-value = 2.73×10^{-131}), which we confirmed using *piaB* and *lytA*-specific qPCR assays. We found that *Streptococcus* (14) presence was highly predictive of pneumococcal carriage in children (<10 years old; positive predictive value [PPV] ranging from 0.78 to 0.81) but lower in those ≥10 years old (PPV 0.21 to 0.56) (Figure S5B; X^2 test p values < 0.05). Furthermore, we observed a strong correlation between the absolute abundance of *Streptococcus* (14) and *piaB* and *lytA* concentration (Figures S5C and S5D; Pearson r = 0.78 and r = 0.75, respectively; p values < 0.001). In line with the results for *Streptococcus* (14), we detected a statistically significant positive association between *S. pneumoniae* and recent mild upper respiratory infection symptoms (with and without fever) in infants (Figure S5E). Concomitantly, important commensals decreased in those suffering from respiratory symptoms, including *Corynebacterium* (7) and *D. pigrum* (3) (Figure 6C; q value < 0.05). Age-stratified analysis showed that these differences were most apparent in infants (0 year-olds) and children (1–14 year-olds) (Figure 6E; q value < 0.05). Compared with the nasopharynx, a smaller degree of variation in the salivary and oropharyngeal microbiota could be attributed to respiratory symptoms (R^2 = 0.7% and 0.1%, respectively; p values < 0.05). For the salivary microbiota (< 10 years), we observed a higher microbial diversity (Figure 6A; q value = 0.002) and higher absolute abundance of 13 genera within individuals with respiratory symptoms and fever when compared with participants without symptoms. (Figure S5A; q values < 0.25).

We also investigated the relationship between the occurrence of pneumonia and the URT microbiota in individuals aged 15 years or older (data not available for under 15 year-olds), correcting for antibiotics use within the last three months (Pearson r = 0.15; q value < 0.001). Of the participants, 5.6% (N = 122) had experienced one case of pneumonia within the past three years, while 2.7% (N = 59) reported multiple cases of pneumonia within this time period. In the nasopharynx of individuals who had experienced multiple pneumonia episodes, the absolute abundance and presence of *Streptococcus* (14) were higher compared with individuals who did not suffer from pneumonia (Figures 6F and 6G; q values < 0.05). These findings aligned with a modest association between *S. pneumoniae* (*piaB* concentration) and multiple pneumonia episodes in the past 3 years (Figure S5E; p value = 0.20), again suggesting that *Streptococcus* (14) and its association with past infection burden reflect pneumococcal carriage in at least part of these individuals. Furthermore, through a sensitivity analysis, we demonstrated that history of pneumonia was associated with higher *S. pneumoniae* abundance, regardless of whether an individual had received antibiotic treatment. In the oropharynx, Shannon diversity decreased in a stepwise manner in those who experienced one

(G) Percent of participants with *Streptococcus (14)* present (number of reads > 0) in their nasopharynx and oropharynx (≥ 15 years), grouped by history of pneumonia.

*q value ≤ 0.25, **q value ≤ 0.1, ***q value ≤ 0.05. The following reference groups were considered: season of sampling = "summer," respiratory symptoms = "no respiratory symptoms," and pneumonia = "did not suffer from pneumonia".

See also Figure S5.

(q value = 0.02) or recurrent pneumonia episode(s) (q value = 0.005), respectively (Figures S5F–S5H). Similar to what was observed for the nasopharynx, we found a higher *Streptococcus* (14) prevalence in the oropharyngeal microbiota in those with a history of pneumonia (Figure 6G).

DISCUSSION

This nationwide population study provides a comprehensive analysis of the URT microbiota across the full human lifespan, identifying age as the primary driver of the microbiota, with distinct patterns observed for the different URT niches. The nasopharynx of younger individuals exhibits lower diversity and a higher total bacterial density, while the opposite was observed for older individuals. The microbial diversity and total bacterial density stabilized between the ages of 15 and 24, suggesting that the nasopharyngeal microbiota continues to develop and mature into early adulthood. This has not been documented before and is in sharp contrast with the development of the gut microbiota, which already stabilizes around the age of 2–5 years.[24] The absolute abundance of *D. pigrum* and *Corynebacterium spp.*, essential commensals of the nasopharynx, has been frequently linked with the absence of disease[2] and inhibition of potential pathogens.[25] We found that the abundance of these bacteria varied across age, with *D. pigrum* being more abundant in younger individuals yet commonly present across all age categories. Additionally, we uncovered a consortium of rare taxa, including *Anaerococcus (octavius)*, *Corynebacterium*, *Finegoldia magna*, *Lawsonella clevelandensis*, and *Peptoniphilus* that characterized the adolescent and adult nasopharyngeal profile ($\geq$15 years). These taxa were more abundant in males, especially during adolescence, and these patterns remained consistent when stratifying for other factors such as smoking behavior. We speculate that the combined age- and sex-driven variation in microbiota composition may be related to hormonal fluctuations, which have previously been associated with changes in the gut microbiota composition in mice.[26] In addition, we hypothesize that microbiota development across age groups may be (reciprocally) linked to age-driven immune development,[27] epithelial changes,[28] anatomical development,[29] and differences in environmental exposures across the lifespan. Compared with the nasopharynx, the oral niches had a higher bacterial density and diversity while exhibiting lower variability between individuals and more subtle age-related differences, which might result from its role in food digestion.[30]

We observed clear niche- and age-specific associations between microbial community composition and host and environmental characteristics, among others indicating that social contact is a strong driver of the nasopharyngeal microbiota composition. The nasopharynx of adults living or working with children more closely resembled the typical nasopharyngeal microbiota found in children, with a higher absolute abundance of *Moraxella*. Although previous studies found that the nasopharynx of infants resemble their mothers,[31] our data suggest that young children may also influence their parents' microbiome.

Recent respiratory symptoms, antibiotic use, and active smoking were negatively associated with the absolute abundance of the nasopharyngeal commensals *D. pigrum* and *Corynebacterium*. Interestingly, the absolute abundance of streptococci, confirmed to be largely representing *S. pneumoniae*, and *Haemophilus* spp. was related to recent mild infections, including upper airway infections in children, and lower respiratory infections, including pneumonia, in adolescents/adults. These findings were most apparent in infants and children and add to the expanding body of literature suggesting that bacterial communities in the nasopharynx may modulate the severity of viral infections as well as require a phase of recovery before eubiosis is re-established.[2,32]

Our data suggest that the oral microbiota is mainly driven by lifestyle factors such as tobacco and alcohol use. Among others, we found lower microbial diversity in active and past active smoking, emphasizing the influence of smoking on the oral microbiota. Recent evidence suggests a link between cardiovascular disease and the oral microbiota, in addition to the well-established connection with the gut microbiota.[5] At this stage, however, further studies are needed to elucidate changes in the oral microbiota as a direct or mediatory factor that contributes to the risk of cardiovascular disease.

Previous studies of the microbiota across various URT niches have been limited in scope, primarily focusing on either small sample sizes or specific stages within the human lifespan.[33–41] We shed light on age-related dynamics of the URT microbiota in a general population, filling a knowledge gap that has persisted despite numerous investigations of the human microbiome. A major strength of our study is the use of a controlled laboratory setting with standardized operating procedures in place to characterize low-biomass respiratory samples to ensure the reliability of the data. Laboratory procedures were first carefully benchmarked to accommodate low-biomass nasopharyngeal samples.[42] Furthermore, to tackle contamination and avoid using low-quality samples, we employed extensive pre-processing steps.

Understanding the potential transmission of microbes through the environment is essential for effective disease prevention and management. Therefore, the impact of environmental factors, such as air pollution and intensive livestock farming,[43,44] is an area of research that warrants further investigation. Overall, we here comprehensively describe the URT microbiota across different age categories in a large open population study, highlighting the most important niche- and age-specific factors that impact the microbial community composition of the URT. Since the URT microbiota is implicated in various short- and long-term respiratory health outcomes, our results may guide future preventive measures to promote respiratory health across the human lifespan and may serve as a baseline for future studies into the URT microbiota.

Limitations of the study

Although our study utilized extensive questionnaires to gather information, including host, environmental, and health characteristics, we did not collect specific data on oral hygiene practices such as tooth brushing frequency or the use of oral washes.[45] This gap in information could potentially impact our understanding of how oral hygiene habits influence the composition of the URT microbiota, especially for the salivary and oropharyngeal microbiota. Additionally, the reliance on self-reported data through questionnaires introduces the possibility of recall bias, where participants may inaccurately remember or report information about

their medical history. Our study employed a cross-sectional design, which limits our ability to establish causality or determine the temporal relationship between variables. Our samples were collected at varying consultation hours, which could introduce variability due to factors such as time of day and potential differences in microbial profiles associated with these variations.[46] Furthermore, 16S-rRNA amplicon sequencing, the method we used for microbiome profiling, has inherent limitations. This technique may not differentiate closely related species within genera due to the conserved nature of the 16S-rRNA gene and does not allow for functional readouts. Future studies could include metagenomic approaches to address these limitations. Additionally, to capture the full breadth of normal variation in the URT microbiota, future research should include longitudinal studies conducted across diverse geographical locations and lifestyles.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.cell.2024.07.008.

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AUTHOR CONTRIBUTIONS

M.-L.O. pre-processed, analyzed, and visualized the data. M.-L.O., W.A.A.d.S.P., E.F., R.P., L.A.M.S., R.M., K.T., and D.B. assisted in the statistical analyses and interpretation of the results and were responsible for the writing of the original draft. E.F., F.R.M.v.d.K., E.A.M.S., D.B., and T.B. were responsible for conceptualizing the study and acquiring funding. M.L.J.N.C., J.A.G., E.M.v.L., R.H., S.K., and R.M. were responsible for the execution and quality control of the laboratory work. All authors contributed to interpretation of the results, critically revised the manuscript for important intellectual content, and approved the final manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Bacterial and virus strains		
ZymoBIOMICS microbial community standard	Zymo Research, CA, USA	Cat#D6300
ZymoBIOMICS microbial community DNA standard	Zymo Research, CA, USA	Cat#D6306
Biological samples		
Nasopharynx, oropharynx and saliva samples	Collected primarily from neighborhood community center, the Municipal Health Service (GGD) or Saltro testing location.	third seroepidemiological study in the Netherlands (PIENTER3)
COPAN eSwab, 482CE (transoral nasopharyngeal swab)	Copan Diagnostics Inc., CA, USA	Cat#482CE
COPAN eSwab, 484CE (transnasal nasopharyngeal swab)	Copan Diagnostics Inc., CA, USA	Cat#484CE
COPAN eSwab, 480CE (oropharyngeal swab)	Copan Diagnostics Inc., CA, USA	Cat#490CE
Oracol saliva collection system	S10 Malvern Medical	N/A
Chemicals, peptides, and recombinant proteins		
Zirconium beads (0.1mm)	Biospec Products, OK, USA	Cat#11079101z
Phenol	VWR, PA, USA	Cat#108-95-2
Master mix universal taqman 5×5 mL	Thermo Fischer Scientific, MA, USA	Cat#10437304
HPLC grade water	Instruchemie, The Netherlands	Cat#2195
10mM dNTP mix	Roche, Switzerland	Cat#11814362001
phiX control v3	Illumina, CA, USA	Cat#FC-110-3001
Critical commercial assays		
Mini-Beadbeater-24	Biospec Products, OK, USA	Cat#112011EUR
StepOnePlus RealTime PCR System	Thermo Fisher Scientific, MA, USA	Cat#4376600
LightCycler 480 PCR System	Roche, The Netherlands	N/A
Mag Mini DNA extraction kit	Immunosource, Belgium	Cat#NAP40401
Phusion Hot Start II High-Fidelity DNA Polymerase	Thermo Fisher Scientific, MA, USA	Cat#F-549L
Quant-iT PicoGreen dsDNA Assay Kit	Thermo Fisher Scientific, MA, USA	Cat#P7589
Beckman Coulter Agencourt AMPure XP	Thermo Fisher Scientific, MA, USA	Cat#A63880
MiSeq Reagent kit V3 (2x300 bp)	Illumina, CA, USA	Cat#MS-102-3003
Illumina MiSeq instrument	Illumina, CA, USA	Cat#SY-410-1003
Deposited data		
Raw 16S-rRNA sequencing data	This paper; NCBI SRA	Bioproject: PRJNA997934
Silva v138 (Version 2; August 2020)	Quast et al.[47]	Zenodo: https://zenodo.org/record/3986799#.YfD5ti-iH0r
Source code/scripts to process and analyse the data	This paper	GitLab: https://gitlab.com/Mari-Lee/URT_microbiota_baseline; Zenodo: https://doi.org/10.5281/zenodo.10854139
Oligonucleotides		
forward primer 16S-F1 5'-CGA AAG CGT GGG GAG CAA A-3'	Bogaert et al.[48]	N/A
reverse primer 16S-R1 5'-GTT CGT ACT CCC CAG GCG G-3'	Bogaert et al.[48]	N/A

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
probe 16S-P1 FAM-ATT AGA TAC CCT GGT AGT CCA-ZEN	Bogaert et al.[48]	N/A
515F 16S V4 forward primer 5'-GTGCCAGC MGCCGCGGTAA-3' (including Illumina adapters and barcodes)	Caporaso et al.[49]	N/A
806R 16S V4 reverse primer 5'-GGACTACHVGGGTWTCTAAT-3' (including Illumina adapters and barcodes)	Caporaso et al.[49]	N/A
Forward primer *piaB* 5'-CAT TGG TGG CTT AGT AAG TGC AA-3'	Trzciński et al.[50]	N/A
Reverse primer *piaB* 5'-TAC TAA CAC AAG TTC CTGA TAA GGC AAG T-3'	Trzciński et al.[50]	N/A
Probe *piaB* 5'-FAM-TGT AAG CGG AAA AGC AGG CCT TAC CC-3'-TAMRA	Trzciński et al.[50]	N/A
Forward primer *lytA* 5'-ACG CAA TCT AGC AGA TGA AGC A-3'	Carvalho et al.[51]	N/A
Reverse primer *lytA* 5'-TCG TGC GTT TTA ATT CCA GCT-3'	Carvalho et al.[51]	N/A
probe *lytA* 5'-FAM-GCC GAA AAC GCT TGA TAC AGG GAG-3'-BHQ52	Carvalho et al.[51]	N/A
Software and algorithms		
R Statistical Software v4.3.2	R Core Team	https://www.r-project.org
RStudio Server v2023.3.0.386	RStudio	https://posit.co
Adobe Illustrator v28.3	Adobe	https://www.adobe.com/products/illustrator.html
DADA2 v1.26	Callahan et al.[52]	https://benjjneb.github.io/dada2/
snakemake v5.18.1	Mölder et al.[53]	https://snakemake.readthedocs.io/en/stable/
decontam v1.18.0	Davis et al.[54]	https://bioconductor.org/packages/release/bioc/html/decontam.html
phyloseq v1.45.0	McMurdie and Holmes[55]	https://joey711.github.io/phyloseq/
vegan v2.6-4	Oksanen et al.[56]	https://cran.r-project.org/web/packages/vegan/index.html
MaAsLin2 v1.15.1	Mallick et al.[57]	https://bioconductor.org/packages/release/bioc/html/Maaslin2.html
segmented v2.0-3	Muggeo et al.[58]	https://cran.r-project.org/web/packages/segmented/

RESOURCE AVAILABILITY

Lead contact

Further information and requests for reagents may be directed to the lead contact Debby Bogaert (D.Bogaert@ed.ac.uk).

Materials availability

All samples generated in this study are stored at the Centre for Infectious Diseases Research, Diagnostics and Laboratory Surveillance of the National Institute for Public Health and the Environment (RIVM). Samples are not part of an open biorepository due to the terms of the informed consent. Requests for resource sharing can be made to the corresponding author and will be evaluated on a case-by-case basis.

Data and code availability

- The paired FASTQ files containing 16S-rRNA sequencing data have been submitted to the NCBI Sequence Read Archive (SRA) database and are publicly accessible as of the publication date. The accession numbers can be found in the key resources table.

- Code used to process and analyse the data is available at Gitlab: https://gitlab.com/Mari-Lee/URT_microbiota_baseline. A version of the code has been archived in a Zenodo repository and is accessible to the public as of the publication date. DOIs are listed in the key resources table.
- Any additional information required to reanalyse the data reported in this paper is available from the lead contact upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Study population and design

During phase I of the TRIuMPH project, the technical capacity building of microbiota research at the National Institute for Public Health and the Environment of the Netherlands (RIVM) was realised and a microbiota dataset of a nationwide cohort was generated. Upper respiratory tract (URT) samples were collected in the context of a large cross-sectional population-wide study (PIENTER-3, 2016/2017), previously described.[19] PIENTER-3 is the third seroepidemiological study executed by the RIVM and is used to monitor the seroprevalence of National Immunisation Programme (NIP)-targeted diseases. A two-phase cluster sampling method was employed to select a demographic cross-section of Dutch residents aged 0 to 89 years. Forty municipalities were selected from five regions in proportion to their size. Within each chosen municipality, a sample was drawn from the population register, stratified by age, with 89 years as the maximum age for inclusion. Consequently, age groups were delineated as 0, 1–4, 5–9, and so forth, up to 75–79 and 80–89 years of age. Additionally, this cohort deliberately included an overrepresentation of non-Western migrants, individuals residing in areas with low vaccination coverage, and an additional sample of individuals born in Suriname, Aruba, and the former Dutch Antilles. Participants in the study provided various data during consultation hours, including questionnaires, blood samples, oro- and nasopharyngeal swabs, fecal samples, and saliva samples, while also maintaining contact pattern diaries. Additionally, vaccination records and medical histories were collected as part of the dataset. Information about participant's sex was self-reported and gathered through the questionnaires. Of the participants, 51% were female (Table 1; Table S1).

Material collection

Deep phenotypic data was collected by four questionnaires directed at 0-14 year-olds and participants 15 years of age and older (two initial and two follow-up questionnaires). Questionnaires included data on socio-demographics, vaccination, health and disease status, behaviour and diet. All participants filled out the initial questionnaires, while the response rate for the follow-up questionnaires was 73%. To form our study population, nasopharyngeal swabs, oropharyngeal swabs and salivary samples were collected during consultation hours. Because it is technically difficult to obtain a pure oropharyngeal sample (backwall oral cavity) in young children, we collected saliva instead, as this is non-invasive, easy to collect, reliable, and in young children, has shown high similarity with oropharyngeal samples.[59] Saliva samples were obtained using a sponge swab, which was gently positioned between the cheek and jaw, allowing to absorb sufficient saliva for one minute. Following, saliva was promptly extracted from the sponge by squeezing into a cryovial tube. These tubes were immediately placed on dry ice at the time of collection and stored at −80 °C. Obtaining samples from the nasopharynx involved inserting a Copan swab into one of the nostrils, bringing it back along the floor of the nasal passage towards the back wall of the nasopharynx, followed by gentle brushing. In contrast, collecting samples from the oropharynx involved gently inserting a swab into the mouth, directing it toward the back of the throat, while avoiding touching any structures in the oral cavity. Oropharyngeal and nasopharyngeal swabs were placed in liquid Amies medium and directly stored on dry ice, followed by transport and storage at −80 °C.[19]

Informed consent

All procedures performed were in accordance with the ethical standards of the institutional and/or national research committees. The study was approved by the Dutch National Ethics Committee (METC Noord-Holland; M015–022). Written informed consent was obtained from all adult participants and the parents or legal guardians of minors included in the study.

METHOD DETAILS

DNA isolation

Laboratory procedures have been previously described.[42] DNA was extracted from a total of 5,930 URT samples using the Agowa Mag DNA extraction kit (LGC Genomics, Berlin, Germany), following previously established methods with slight adjustments to ensure reliability for extracting low biomass DNA.[60] Each isolation run included a 200 μl aliquot of 10^3 diluted Zymo mock as a positive control, along with two negative controls containing lysis buffer only (referred to as DNA isolation blanks). Samples were thawed on ice and vortexed for 10 seconds. Subsequently, 600 μl of lysis buffer with zirconium beads (diameter 0.1 mm, Biospec Products, Bartlesville, OK, USA) and 550 μl phenol (VWR International, Amsterdam, the Netherlands) were added to a conical 1.5 ml screw-cap Eppendorf tube for each sample. Samples were mechanically disrupted twice using a Mini-Beadbeater-24 (Biospec Products) for 2 minutes at 3,500 oscillations/minute, with a 2-minute transfer on ice in between each bead-beating step. The tubes were then centrifuged for 10 minutes at 4500×g. The clear aqueous phase was transferred to a 2 ml Eppendorf tube containing 1.3 ml binding buffer and 10 μl magnetic beads. After shaking for 30 minutes, the tubes were placed in a magnetic separation rack. The supernatant

was discarded, and the magnetic beads were washed with wash buffers 1 and 2, followed by air-drying for 15 minutes at 55°C. DNA was eluted in either 35 μl or 50 μl elution buffer, depending on the starting material, by shaking for 15 minutes at 55°C. The supernatant containing the eluted DNA was transferred to a 1.5 ml Eppendorf LoBind tube and stored at -20°C.

16S-rRNA sequencing

Following the standard protocol, the hypervariable region 4 (V4) of the 16S-rRNA-gene was amplified using PCR and the 515F/806R-primer pair including the Illumina adapters and sample specific barcodes.[49] Each 25 μl PCR reaction was composed of 0.5 μl Phusion Hot Start II High-Fidelity DNA Polymerase, 5 μl 5× Phusion HF Buffer (Thermo Fisher Scientific, MA, USA), 7 μl HPLC grade water (Instruchemie, Delfzijl, the Netherlands), 2.5 μl of 2 mM dNTP mix (Roche, Mannheim, Germany), 5 μl of 5 μM barcoded primer 515F and barcoded primer 806R, and 5 μl template DNA. PCR reactions followed a specific protocol: initial denaturation at 98 °C for 30 seconds; 30 cycles at 98 °C for 10 seconds, 55 °C for 30 seconds, and 72 °C for 30 seconds; and a final extension at 72 °C for 5 minutes. Samples with a 16S rRNA gene DNA concentration of less than 20 pg/μl (corresponding to <100 pg input DNA) were used undiluted, while those with higher concentrations were appropriately diluted in HPLC grade water. The fragment size of the amplicon was assessed through agarose gel electrophoresis and quantified using the Quant-iT PicoGreen dsDNA Assay Kit (Thermo Fisher Scientific, MA, USA). Barcoded amplicons were pooled in equimolar ratios. A total of 23 amplicon pools (5-27) were purified, implementing two consecutive purifications using 0.9× AMPure XP magnetic beads. Four pools (1-4) were purified using a combination of agarose gel and 0.9× AMPure XP magnetic beads. The purified pools were sequenced using MiSeq reagent kits v3. The samples were sequenced in 27 Illumina MiSeq runs (Illumina Inc., San Diego, CA, US). Additionally, both positive and negative controls were incorporated to validate the sequencing process and identify potential contamination. Technical controls included DNA isolation blanks (n = 351), sequencing blanks (n = 80), Zymo mock bacteria (ZMCB, n = 189) and Zymo mock DNA (ZMCD, n = 80). Mock communities were carefully inspected and checked for irregular microbial profiles across the MiSeq-runs.

Bacterial DNA quantification

The bacterial load was quantified using quantitative PCR (qPCR) with the StepOnePlus Real-Time PCR System (Thermo Fisher Scientific, MA, USA) and universal primers and a probe targeting the 16S rRNA gene. The primers and probe used were 16S-F1 (forward), 16S-R1 (reverse), and 16S-P1 (probe). (IDT, Leuven, Belgium).[48,60] To optimise qPCR reproducibility and to allow reliable comparisons of DNA concentrations, we developed a standard curve by using a synthesised fragment of the 16S rRNA gene (gBlocks Gene Fragment, IDT, 5′-CGG TGC GAA AGC GTG GGG AGC AAA CAG GAT TAG ATA CCC TGG TAG TCC ACG CCG TAA ACG ATG TCT ACT AGC TGT TCG TGG TCT TGT ACT GTG AGT AGC GCA GCT AAC GCA CTA AGT AGA CCG CCT GGG GAG TAC GAA CGC AAG-3′).

Taxonomic assignment

Using an in-house bioinformatics pipeline, paired-end reads were processed using DADA2[52] (v1.16.0; maxEE = 2; truncLen = 200/150) as reported before,[11] allowing the inference of amplicon sequence variants (ASVs). Chimeras were detected and eliminated using the 'consensus' method. Taxonomy was assigned using the naïve Bayesian classifier and the Silva v138 (Version 2; August 2020) reference database.[47] Implementation of DADA2 was managed through the Snakemake workflow management system (version 5.18.1).[53] The output from DADA2 was integrated into a single object using the "*phyloseq*" R-package, which was then utilized for subsequent analyses.[55]

Streptococcus pneumoniae detection with qPCR

Pneumococcal DNA was detected via single-plex qPCR using primers and probes targeting sequences within genes coding for the pneumococcal iron uptake ABC transporter lipoprotein *piaB*[50], and for the major pneumococcal autolysin *lytA*[51] and testing nucleic acids extraction volume equivalent of 1.4 μl within a total qPCR reaction volume of 12.5 μl. The pneumococcal load for *piaB* and *lytA* was quantified using the StepOnePlus Real-Time PCR System (Thermo Fisher Scientific, MA, USA) and the LightCycler 480 PCR System (Roche, Almere, The Netherlands), respectively. Both assays were performed with Taqman Universal Master mix (Thermo Fisher Scientific, MA, USA). A sample was considered positive for *S. pneumoniae* when signals for both targeted genes were of cycle threshold (C_T) <40.[61] Pneumococcal absolute abundances were determined as previously described.[62] In short, a 10-fold serial dilution of *S. pneumoniae* Hungary 19A-6 strain[63] nucleic acids of predetermined DNA concentration was used to construct a standard curve, which allowed us to calculate the (marker-specific) pneumococcal DNA concentration for each given C_T-value.

Data pre-processing

Several filtering steps were conducted to remove contamination and samples of low quality (Figure S6A). Our dataset included 206,867,690 raw sequences in total with an average sequencing depth of 29,696 sequences per sample. Off-target reads were defined as reads outside the normal length range of +3/-3bp (i.e. 250-256bp) and were removed from the dataset.[64] Taxa belonging to the phylogenetic groups 'Mitochondria', 'Chloroplast', 'Archaea' and 'Eukaryota' were removed from the dataset, resulting in a 2% exclusion of ASVs and 0.22% of the reads. We renamed the ASVs using the lowest known taxonomic rank and indicated the rank number (based on mean relative abundance) within brackets.

Using DNA isolation blanks, potential contaminant ASVs were detected separately for the three niches (saliva, oropharynx and nasopharynx) with the 'combined' method of the "*decontam*" R-package ("*isContaminant*" function) using default parameters.[54] In short, the 'combined' method incorporates both the 'prevalence' and 'frequency' methods. The frequency method compares the prevalence of ASVs in samples and DNA isolation controls to discriminate biological signals from contamination. For the frequency method, contaminating ASVs are detected using the inverse relation between DNA density (qPCR) and the relative abundance of ASVs. Using this approach, we identified 18, 213 and 262 potential contaminants in the saliva, oropharynx and nasopharynx samples, respectively, and combined them into a single list. Additionally, we manually identified 17 contaminants that had a relative abundance of more than 1% in at least two DNA isolation controls and an irregular relative abundance across MiSeq runs. Contaminants with a mean relative abundance of more than 0.01% across samples were manually screened, identifying 18 'false positive' hits (i.e. ASVs deemed contaminants by the algorithm yet known to also inhabit the URT microbiota). We specifically focused on ASVs that were identified as contaminants in only one of the three niches. A total of 441 contaminant ASVs were removed from all the samples. These contaminants also had a high relative abundance in the isolation blanks (Figure S6B). Four DNA isolation runs had a high average percentage of contaminant ASVs (>10%) and were removed from our dataset. These runs included samples initially showing low bacterial density after the first isolation, after which we attempted to isolate DNA a second time.

Finally, we excluded four URT samples with a DNA concentration of less than 0.095 pg/μl as determined by the qPCR and 148 URT samples with less than 10,000 reads (Figure S6C), retaining a total of 5,589 URT samples after pre-processing. For downstream analyses, we considered ASVs with a detection rate of 0.1% in at least 2 samples, except when calculating the alpha-diversity indices. An overview of all the subsequent analyses is depicted in Figure S7.

QUANTIFICATION AND STATISTICAL ANALYSIS

All analyses were performed in R v4.3.2 within the Rstudio Server v2023.3.0.386. All statistical tests were two-sided and p values were corrected for multiple testing using Benjamini-Hochberg (BH), unless otherwise specified. We have consistently employed a two-tailed test approach in all our statistical analyses.

Variable selection

We conducted initial and follow-up surveys for both adults (≥15 years old) and children (1-14 years old) in 2016/2017. The initial survey for adults consisted of 67 main questions and the follow-up survey had 22 main questions. For children, the initial survey had 55 main questions and the follow-up survey had 21 main questions. From these four questionnaires, we manually curated 47 factors based on literature and expert consensus on microbiome interactions. These factors were categorised into 4 general, 5 season/symptoms, 6 household/contact, 12 socioeconomic status/lifestyle, 14 medication use/disease, 3 diet and 6 early life factors (Table S1). We evaluated collinearity through Pearson/Phi correlation coefficients and used directed acyclic graphs (DAGs) to identify confounders by visually representing relationships between characteristics. Characteristics with moderate to strong correlations (Pearson *r* between 0.1 to 0.25)[65] were considered for adjustment in our models. We also defined eight age categories (0, 1-4, 5-9, 10-14, 15-24, 25-64 and ≥65 year-old) to identify the non-linear effects of age.

Bacterial density and microbial diversity

For analysing alpha-diversity, we rarefied the count data to normalise for differences in sequence depth (lowest number of reads = 10,040). We used the "*microbiome*" R-package to calculate alpha-diversity indices and assess the microbial diversity of the three niches across ages. We also used a $\log_{10}$ transformation of the bacterial density to visualise the overall bacterial load in the samples. Linear models were used to determine differences between age categories after checking assumptions. We used the age category "40-64 year-olds" as the reference group for the nasopharynx and oropharynx and "1-4 year-olds" as the reference group for saliva. We corrected for multiple testing. Additionally, we used age as a continuous variable within the age groups 5-24 and ≥40 years for the nasopharynx and the ≥15 years age group for the oropharynx. Using the "*segmented*" R-package,[58] we identified breakpoints within these age ranges, which allowed us to assess whether the changes in Shannon index and bacterial density with age were gradual or abrupt. The Davies test was employed to evaluate the significance of the differences in slopes before and after a given breakpoint.

Niche and age specific associations

A Bray-Curtis dissimilarity matrix was generated by implementing the "*vegan*" R-package using the relative abundance of the ASVs.[56] Principal Coordinates Analysis (PCoA) plots were constructed to assess niche and age specific patterns in the URT microbiota composition. Using the "*adonis*" function, a permutational multivariate analysis of variance (PERMANOVA) was conducted to determine both the explained variance (R^2) of niche and age. A total of 365 univariable PERMANOVAs were performed to determine the association between the different host and environmental characteristics and the URT microbiota per age and niche. PERMANOVAs were only conducted when at least 10 participants were in each of the categories under investigation. For this analysis, we did not correct for multiple testing. Homogeneity of variances was checked using permutational multivariate analysis of dispersion with the "*permdisp2*" function.

Most abundant and core genera

To compare the composition of the three niches, the average relative abundance of the most abundant genera (average relative abundance >10%) in individuals aged 7-10 years was determined and visualized in bar plots. Furthermore, we investigated the existence of a core microbiota by identifying the genera present in at least 75% of the samples within the 40-64 year age category for both the nasopharyngeal and oral microbiota. The mean relative abundance of these core genera, along with the residuals (non-core genera), was then visualized across the age categories.

Cluster analysis

Using Bray Curtis distance matrices, complete linkage hierarchical clustering was performed to generate clusters for the nasopharyngeal and oral microbiota of participants with a similar microbial profile. Using the Calinski-Harabasz and Silhouette indices, the optimal number of clusters per niche was determined. Exceptionally large clusters were further stratified. Mosaic plots were used to visualise the cluster membership through the eight age categories. Logistic regression was used to statistically test the association of age as a continuous variable with the cluster membership. We additionally assessed the association between the age categories (0, 1-9, 10-24 and $\geq$65 year-olds) and cluster membership using adults (25-64 years) as reference group. We corrected for the season of sampling and sex.

ASV and genus specific associations

For the nasopharyngeal samples, ASVs with a relative abundance of more than 1% in at least 100 samples and their associations with host and environmental characteristics were further investigated using general linear models using the "*MaAsLin2*" R-package.[57] To reduce the number of features and because we found similar responses in ASVs from the same genera, genus-level was considered for the oral niches. The absolute abundance of the ASVs was calculated by multiplying the qPCR measurements with the relative abundance and used as the outcome variable in our models. For each subset of host and environmental characteristics, we corrected for other variables accordingly.

Sensitivity analyses

To ascertain that the influence of sex on the nasopharyngeal microbiota was not obscured by confounding effects of smoking and drinking status, additional adjustments were made in the multivariable analyses. These adjustments aimed to ensure that observed sex differences weren't solely due to variations in smoking or drinking among those aged 15-24 years. The Akaike Information Criterion (AIC) was used as a measure for model performance, with lower AIC values indicating better balance between model fit and complexity. Comparisons were made between estimates from the original analyses and those from additional adjustments to see if the influence of sex remained consistent across different models. Additionally, we conducted a sensitivity analysis to examine the potential impact of hormonal changes on nasopharyngeal microbiota. Comparing females likely undergoing menopause (60-74 years) and younger females (30-44 years) with males from corresponding age groups, we focused on non-smokers to minimize confounding effects.

Detecting local alignments

We utilised the Basic Local Alignment Search Tool (BLAST) and the 16S ribosomal rRNA database to obtain a more specific taxonomic rank for the most abundant ASVs (those with a relative abundance of greater than 1% in at least 100 samples) by using default parameters and requiring 100% identity.

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Supplemental figures

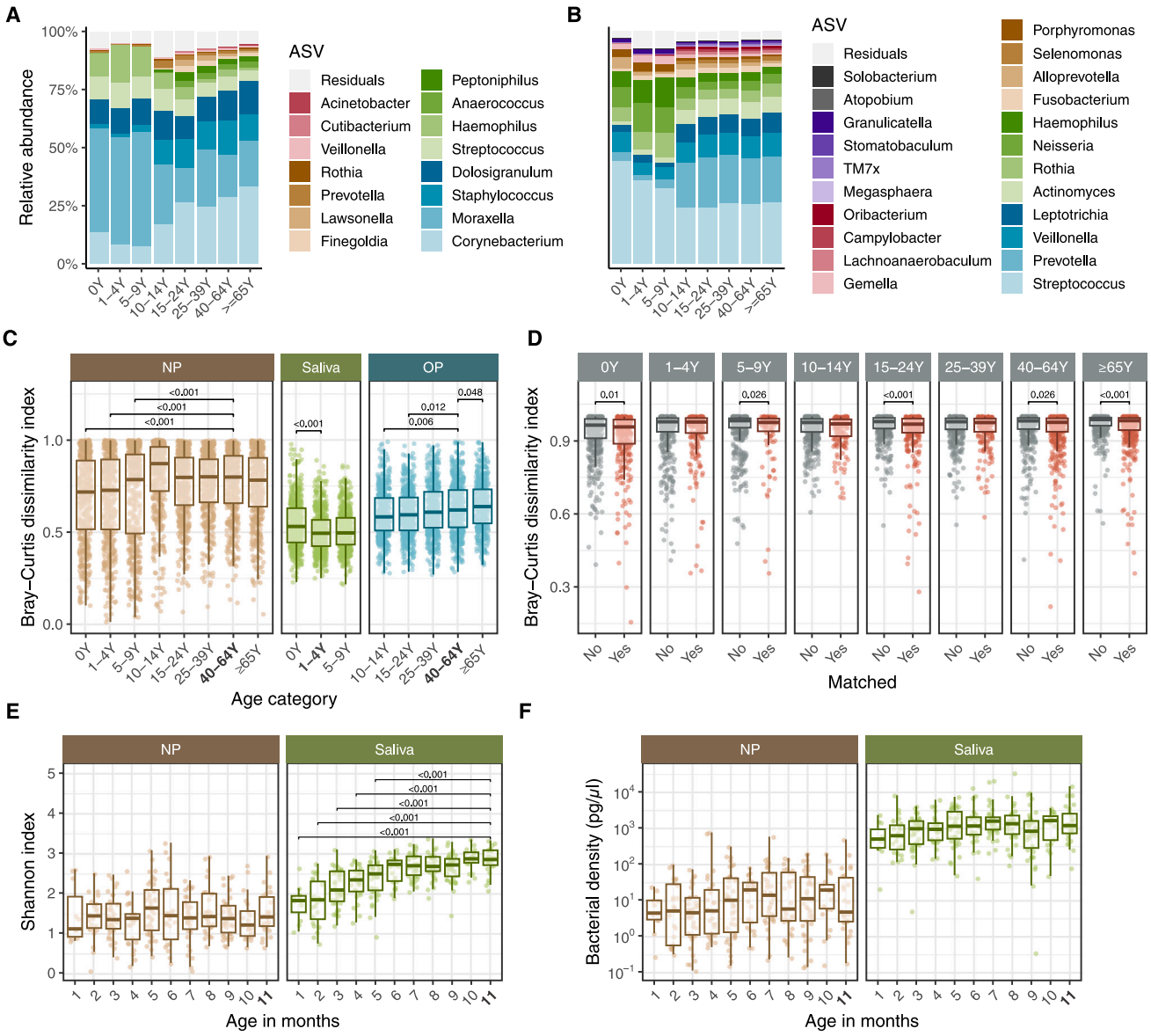


Figure S1. The upper respiratory tract microbiota across ages, related to Figure 2

(A and B) The core genera (present in at least 75% of the samples) identified within the 40–64 years age category for the nasopharyngeal (A) and oral (B) samples. The mean relative abundance of the core genera and residuals (non-core genera) were visualized across age category.

(C) Bray-Curtis dissimilarities across age, stratified by niche (sub-sampling n = 500). Box plots represent the 25th and 75th percentiles (lower and upper boundaries of boxes, respectively), the median (middle horizontal line), and measurements that fall within 1.5 times the interquartile range (IQR; distance between the 25th and 75th percentiles; whiskers). p values were calculated using linear models and corrected for multiple testing with the Benjamini-Hochberg (BH) method (q values).

(D) Bray-Curtis dissimilarities compared between matched and unmatched samples per age category. Matched samples included samples from different niches but from the same participant; unmatched samples are samples from different niches and different participants. Box plots represent the 25th and 75th percentiles, the median, and measurements that fall within 1.5 times the interquartile range. p values were calculated using linear models and corrected for multiple testing with the BH method (q values).

(E and F) ASV-level Shannon diversity (rarefied) (E) and bacterial density (F) in infants across age in months, stratified by niche. Box plots represent the 25th and 75th percentiles, the median, and measurements that fall within 1.5 times the interquartile range. p values were calculated using linear models and corrected for multiple testing with the BH method (q values). Linear regression models were adjusted for sex, season of sampling, birth mode, and breastfeeding.

The reference groups are indicated in bold.

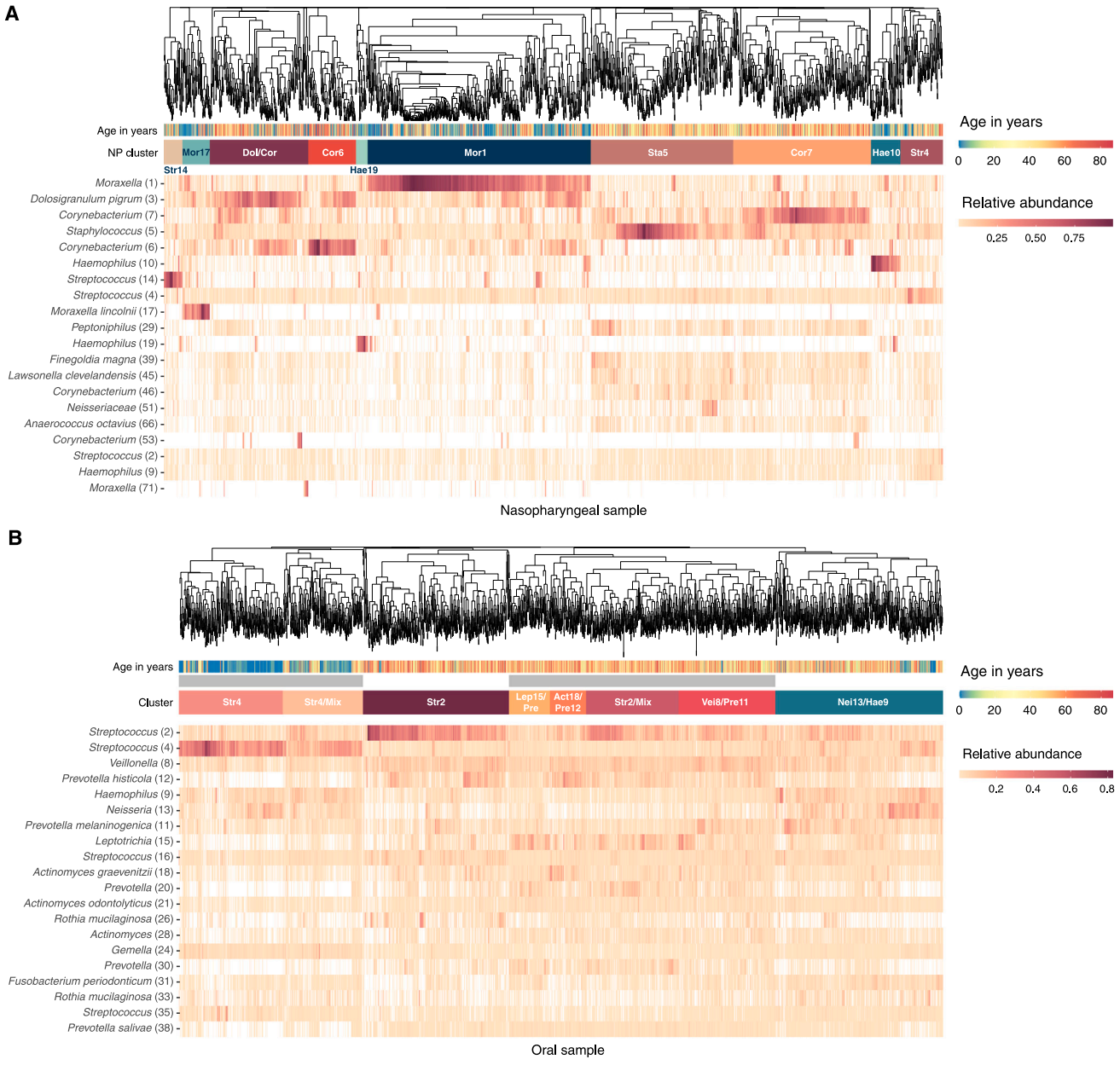


Figure S2. Complete linkage hierarchical clustering of nasopharyngeal (*n* = 2,485) and oral (*n* = 3,104) samples based on the Bray-Curtis dissimilarity matrix, related to Figure 3

The similarities between the samples are depicted by the length of the branches in the tree structure. Adjacent to the branch ends is information on (1) age in years of the participants, (2) clustering resulting in a total of ten nasopharyngeal (A) and eight oral (B) clusters. The cluster names are based on the most discriminative ASVs within each cluster as determined by MaAsLin2 (Table S4; ASVs present at 1% relative abundance in at least 100 samples). The heatmap shows the relative abundance of the 20 highest-ranked ASVs based on the mean relative abundance across all nasopharyngeal and oral samples. We included (supervised) sub-clustering of the two largest clusters from the four initial oral clusters indicated in gray.

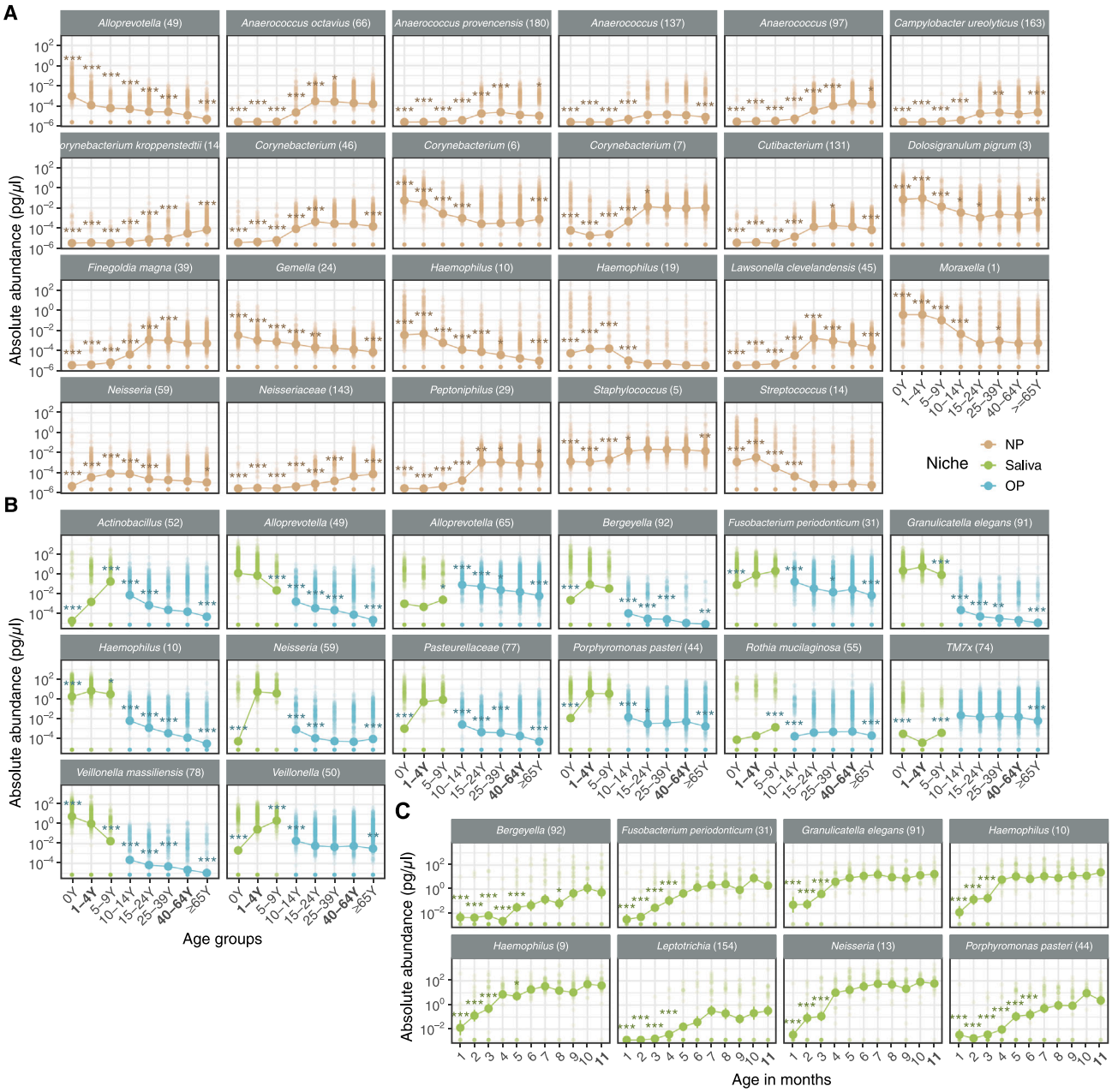
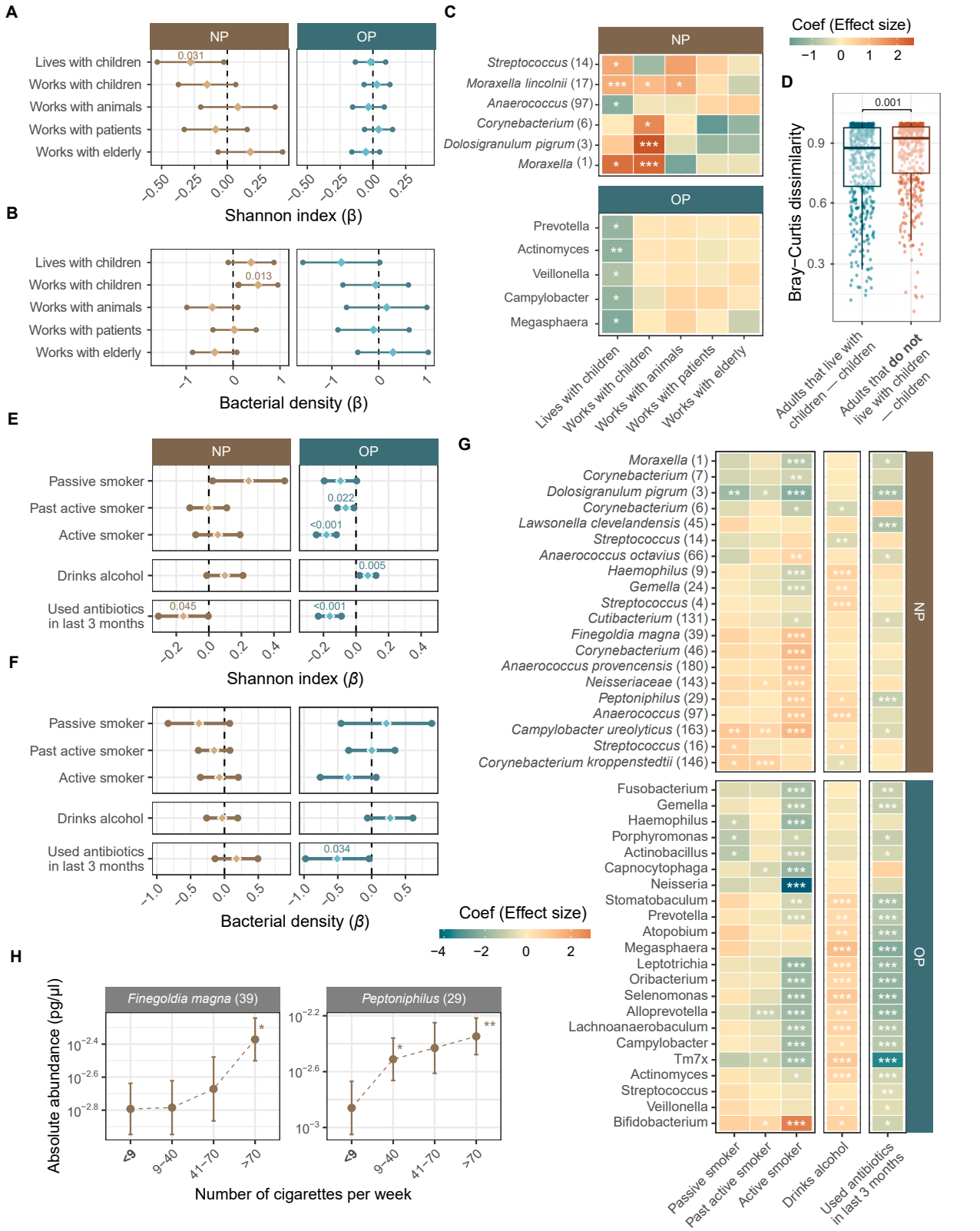


Figure S3. Age-related patterns of the most abundant ASVs in the upper respiratory tract microbiota, related to Figure 3

(A and B) Absolute abundance across age of the ASVs with a significant response to age in at least 3 contrasts (coef < −1 or coef > 1; *q* value < 0.05). Estimates and *q* values were calculated using MaAsLin2 for the nasopharyngeal (ASVs present at 1% relative abundance in at least 100 samples, *n* = 31) (A) and oral (ASVs present at 2% relative abundance in at least 100 samples, *n* = 47) (B) microbiota (related to Table S5). Plots visualize the mean and standard error. The following reference groups were considered: 40–64 years for the nasopharynx and oropharynx and 1–4 years for saliva. All models were adjusted for sex and season of sampling.

(C) Age-related patterns of the most abundant ASVs in the upper respiratory tract microbiota of infants (coef < −1 or coef > 1; *q* value < 0.05). Estimates and *q* values were calculated using MaAsLin2 for the nasopharyngeal (ASVs present at 1% relative abundance in at least 100 samples, *n* = 17) and the oral (ASVs present at 2% relative abundance in at least 100 samples, *n* = 27) microbiota (related to Table S5). Plots visualize the mean and standard error. The reference group was 11 month year-olds. The models were adjusted for sex, season of sampling, birth mode, and breastfeeding.

*q value ≤ 0.25, **q value ≤ 0.1, ***q value ≤ 0.05.



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Figure S4. Social contact and lifestyle-related patterns in the upper respiratory tract microbiota, related to Figure 4

(A–D) Social contact-related patterns in adults (24–64 years).

(A and B) Relationship between Shannon diversity (A) and bacterial density (B) (outcome) with social contact (predictors), stratified by niche. The horizontal lines connecting the circles illustrates the lower and upper bounds of the 95% confidence interval. *p* values were calculated using linear models and corrected for multiple testing with the Benjamini-Hochberg (BH) method (*q* values).

(C) Associations between social contact and the absolute abundance of ASVs (NP) and genera (OP). Estimates and *q* values were calculated using MaAsLin2. The plot shows all the ASVs with at least one significant comparison. The color corresponds to the direction of the association (red for positive, green for negative).

(D) Bray-Curtis dissimilarities compared between the nasopharynx of adults (24–64 years) that live with children (<5 years) versus children (<5 years) and adults that do not live with children versus children. Boxplots represent the 25th and 75th percentiles (lower and upper boundaries of boxes, respectively), the median (middle horizontal line), and measurements that fall within 1.5 times the interquartile range (IQR; distance between the 25th and 75th percentiles; whiskers). *p* value was calculated using linear models.

(E–H) Tobacco, alcohol, and antibiotic use-related patterns in adolescents/adults ($\geq$15 years).

(E and F) Relationship between Shannon diversity (E) and bacterial density (F) (outcome) with tobacco, alcohol, or antibiotic use (predictors). The horizontal lines connecting the circles illustrates the lower and upper bounds of the 95% confidence interval. *p* values were calculated using linear models and corrected for multiple testing with the BH method (*q* values).

(G) Association between tobacco, alcohol, or antibiotic use and the absolute abundance of ASVs (NP) and genera (OP). Estimates and *q* values were calculated using MaAsLin2. The plot shows all the ASVs with at least one significant comparison. The color corresponds to the direction of the association (red for positive, green for negative).

(H) Dose-response relationship of ASVs to weekly cigarette use. The reference group was "<9 cigarettes per week". Estimates and *q* values were calculated using MaAsLin2. Plots visualize the mean and standard error. Asterisks indicate the significance of the concerning group versus the reference group depicted in bold.

*q value $\leq$ 0.25, **q value $\leq$ 0.1, ***q value $\leq$ 0.05. The following reference groups were considered: smoking = "never smokers", drinks alcohol = "non-drinkers", antibiotics use = "did not use antibiotics in the last 3 months". Models for social contact were corrected for age category, sex, season of sampling, lives with children, works with children, works with animals, works with patients, and works with the elderly. Models for smoking and drinking were corrected for age category, sex, season of sampling, level of education, smoking, and drinking. Models for antibiotic use were corrected for age category, sex, and season of sampling.

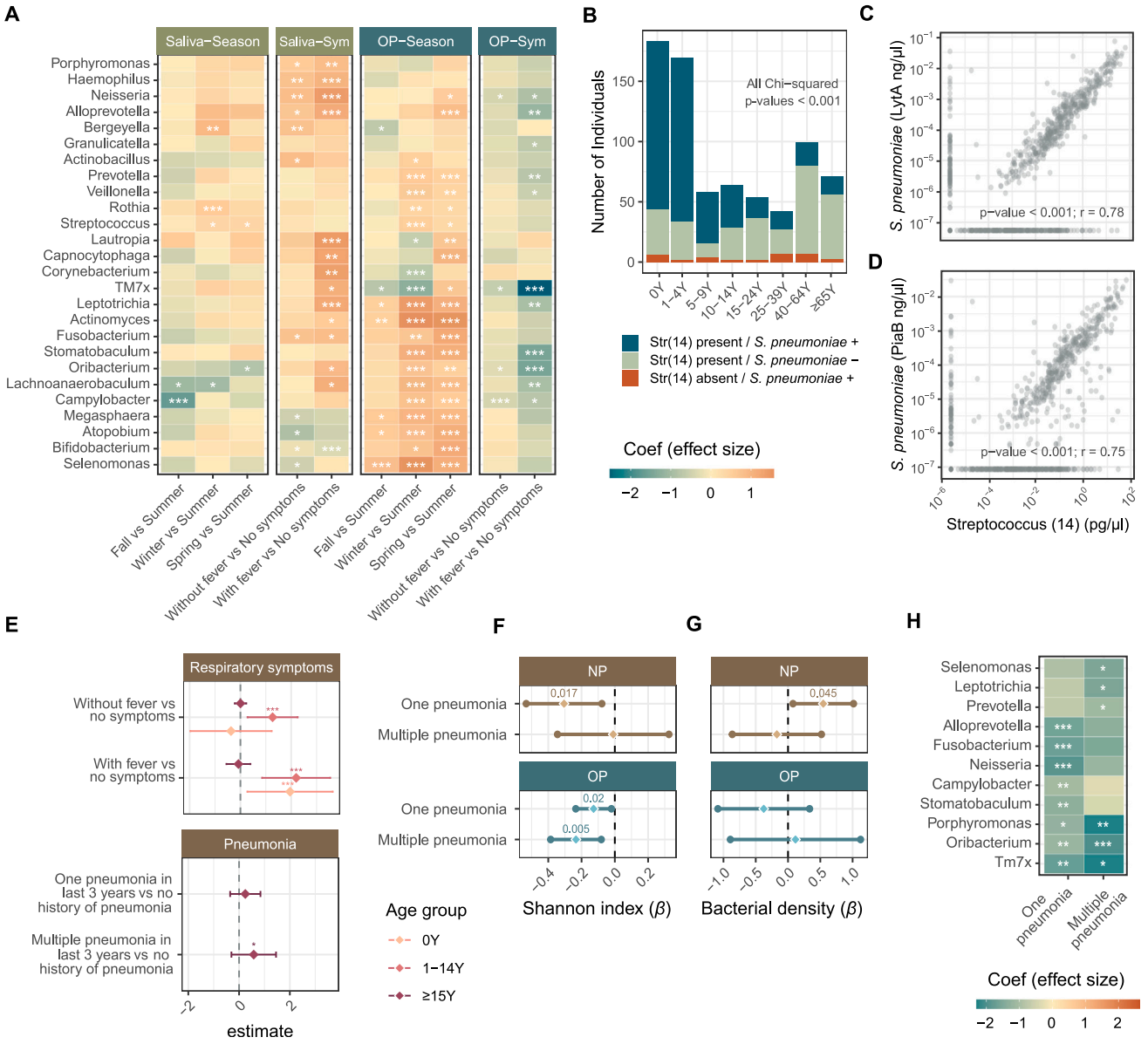


Figure S5. Season, respiratory symptoms, and pneumonia-related patterns in the upper respiratory tract microbiota, related to Figure 6

(A) Association between the absolute abundance of the genera with season of sampling (adjusted for age category and sex) or respiratory symptoms (adjusted for age category, sex, and season of sampling) in the salivary and oropharyngeal (OP) microbiota. Estimates and *q* values were calculated using MaAsLin2. The plot shows all the genera with at least one significant comparison. The color corresponds to the direction of the association (red for positive, green for negative).

(B) Distribution of *Streptococcus* (14) presence/absence (16S-rRNA-based sequencing) and *S. pneumoniae* carriage (*piaB*/*LytA* qPCR) across age categories.

(C and D) Correlation between absolute abundance of *Streptococcus* (14) and *S. pneumoniae lytA* (C) and *piaB* (D) concentration of the nasopharynx (NP). Pearson *r* and *p* values were calculated based on all values. Notably, these strong correlations were specific to *Streptococcus* (14) only, as no significant correlations were observed with other streptococcal ASVs (2, 4, 16, 35, 86, and 94; *p* values > 0.05).

(E) Relationship between $\log_2$-transformed *piaB* concentration (outcome) and respiratory symptoms with/without fever (adjusted for age category, sex, and season of sampling) and pneumonia (one or multiple events in the last 3 years; adjusted for age category, sex, season of sampling, and antibiotic use) in NP stratified by the three age groups 0, 1–14, and ≥15 years. The horizontal lines connecting the circles illustrates the lower and upper bounds of the 95% confidence interval. *p* values were calculated using linear models and corrected for multiple testing with the Benjamini-Hochberg (BH) method (*q* values).

(F and G) Relationship between Shannon diversity (F) and bacterial density (G) (outcome) with pneumonia (predictor) in OP. The horizontal lines connecting the circles illustrates the lower and upper bounds of the 95% confidence interval. *p* values were calculated using linear models and corrected for multiple testing with the BH method (*q* values).

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(H) Association between the absolute abundance of the genera and history of pneumonia in OP (≥ 15 years) (adjusted for age category, sex, season of sampling, and antibiotic use). Estimates and *q* values were calculated using MaAsLin2. The plot shows all the genera with at least one significant comparison. The color corresponds to the direction of the association (red for positive, green for negative).

*q value $\leq$ 0.25, **q value $\leq$ 0.1, ***q value $\leq$ 0.05. The following reference groups were considered: season of sampling = "summer", respiratory symptoms = "no respiratory symptoms", and pneumonia = "did not suffer from pneumonia".

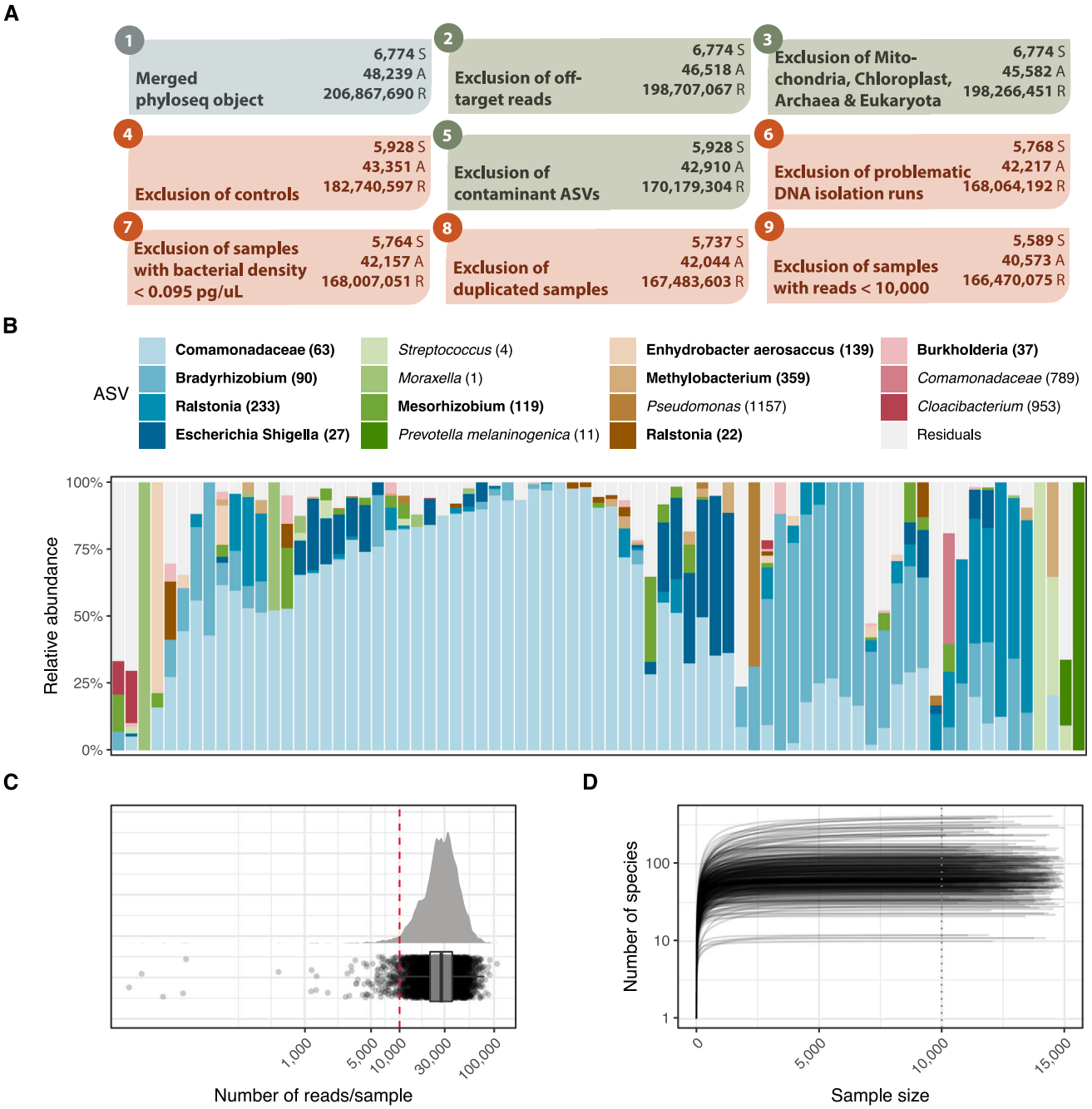


Figure S6. Overview of all quality controls and filtering steps used to remove contamination and samples of low quality, related to STAR Methods

(A) The number of samples (S), ASVs (A), and reads (R) after each filtering step.

(B) Relative abundance of the top 15 ASVs in the isolation blanks. Bold indicates the ASVs that were identified as contaminants and were removed from the dataset.

(C) Raincloud plot of the distribution of the number of reads per sample. Samples with a read count of ≥10,000 reads were retained in our dataset.

(D) Rarefaction curves for samples with <15,000 reads, showing that the microbial diversity has been adequately captured through the sequencing process.

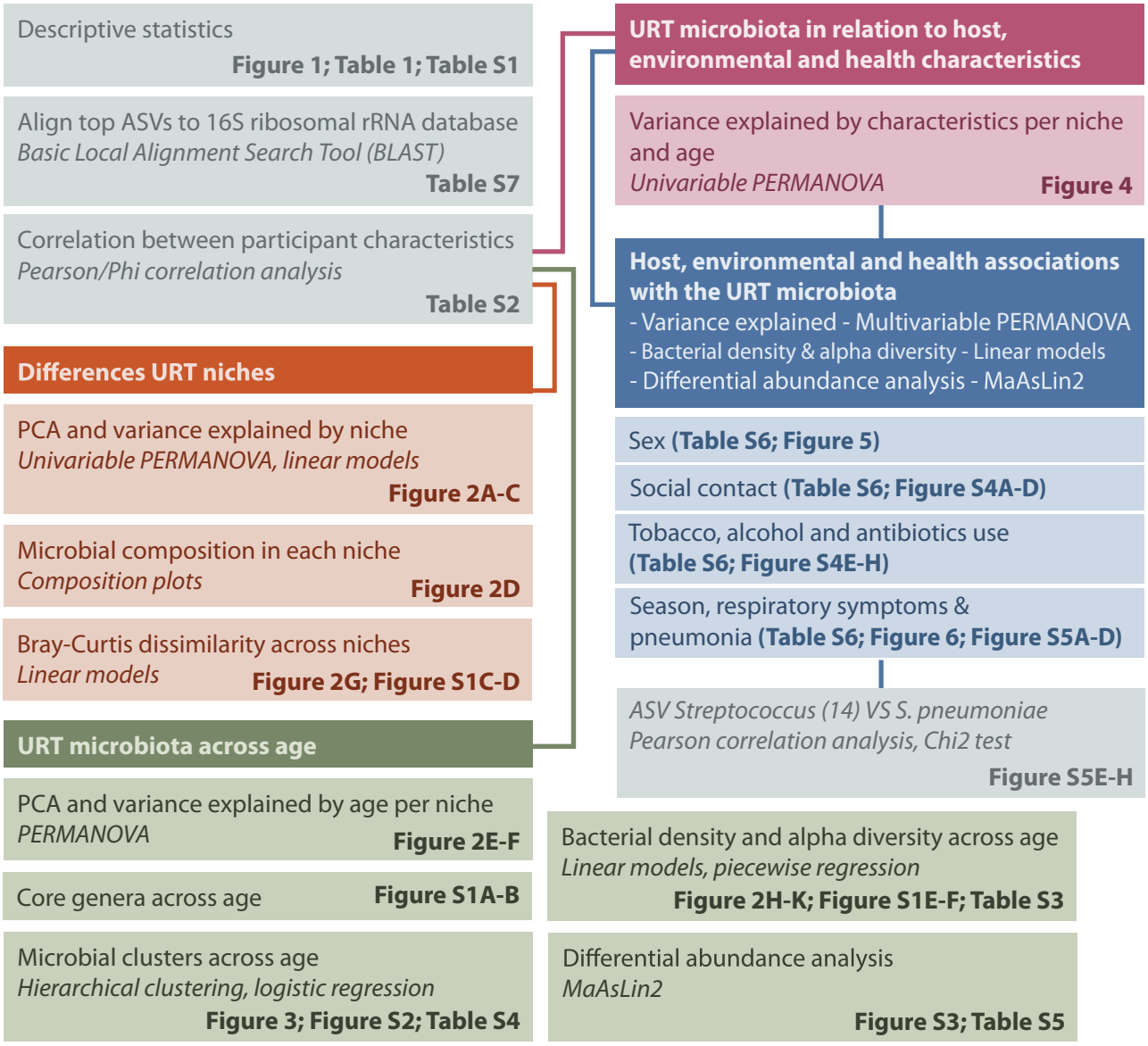


Figure S7. Flowchart of all data analyses, related to STAR Methods

Overview of all statistical analyses utilized, along with the corresponding figures/tables that display the results of these analyses. A solid line is used to illustrate the connections between the different analyses, including any nested analyses.